



Nicholls State University

NICHOLLS STATE UNIVERSITY

HUMAN SUBJECTS INSTITUTIONAL REVIEW BOARD

REQUEST FOR EXEMPT REVIEW BY HUMAN SUBJECTS INSTITUTIONAL BOARD

Nicholls State University has established standards and guidelines to ensure adequate protection is provided to individuals participating in a research activity. The Human Subjects Institutional Review Board (HSIRB) is charged with the responsibility of screening all research which employs human participants conducted by faculty, administrators, or students affiliated with Nicholls State University. The guidelines employed for screening are those set forth by university policy. Please fill in all requested information and keep a copy of this form and any supporting documentation on file.

General Guidelines:

- 1. Primary Investigator:** The primary investigator planning a research activity involving human subjects should obtain a Request for HSIRB application form and complete all requested fields fully with authentic, descriptive details. Research originating from other institutions should be approved by the host institution prior to applying for approval at Nicholls State University.
- 2. Review Workflow:** The completed application is submitted through the PRAMS platform. An initial review of the application will be made by the college HSIRB representative to determine if the project is considered Category I, EXEMPT, Category II, EXPEDITED REVIEW, or Category III, FULL COMMITTEE REVIEW.
- 3. Timing:** Investigators must receive formal approved status from the board BEFORE any recruitment actions or data collection commence. Retroactive approvals are strictly prohibited.



Nicholls State University

NICHOLLS STATE UNIVERSITY
HUMAN SUBJECTS INSTITUTIONAL REVIEW BOARD
PROJECT INFORMATION FORM

Title of Investigation:	A Study in Decision Making
Name of Primary Investigator:	Dr. Christopher Castille
PI Title / Academic Rank:	Associate Professor of Management
PI Department / School:	Management and Marketing
PI Email Address:	christopher.castille@nicholls.edu
PI Phone Number:	985-449-7015
Other Investigators Involved:	Dr. Ann-Marie R. Castille (Associate Professor of Management), ann-marie.castille@nicholls.edu, 985-448-4738 Dr. Samantha Falgout (Assistant Professor of Accounting), samantha.falgout@nicholls.edu, 985-448-4193 Dr. Kaitlin Gravois (Assistant Professor of Marketing), kaitlin.gravois@nicholls.edu, 985-448-4187 Dr. Adrien Maught (Assistant Professor of Marketing), adrien.maught@nicholls.edu, 985-448-4194
Source of Project Funds:	Nicholls State University Research Council
Is this a Continuation of research?	No
Federal Exemption Category:	Exempt Category 2 (45 CFR 46.104(d)(2))

1. Description of Project or Proposal

This study is a local replication of the foundational Schweitzer, Ordóñez, and Douma (2004) experiment examining whether performance goals drive unethical behavior. Participants complete anagram puzzles under one of four conditions: do-your-best, mere goal, reward goal, or personal goal. This design directly addresses several methodological limitations of the original 2004 study, tests key boundary conditions—specifically trait loss aversion and goal proximity—and contributes to a multi-site global replication project. Data collection occurs in two phases: a pilot session to calibrate materials and administration procedures, followed by a main study session for hypothesis testing. Both pilot and main materials are supplied for HSIRB review.

2. Population of Human Subjects

Undergraduate students enrolled in Business Administration courses (specifically BSAD 101, meeting in Powell 140) at Nicholls State University, as well as working professionals. All participants must be 18 years of age or older.

3. Research Procedures and Data Collection

Pilot session: (1) Consent using Appendix B pilot form; (2) two 1-minute practice rounds; (3) timed anagram performance rounds under do-your-best instructions; (4) anonymous productivity report (no cash self-payment; participants may use a phone to access an online Scrabble dictionary via a QR code on the report); (5) brief questionnaires; (6) pilot appreciation letter.

Main study: (1) Consent using Appendix B2 main study form; (2) two 1-minute practice rounds; (3) random assignment to do-your-best, mere goal, personal goal, or reward goal; (4) seven 1-minute performance rounds; (5) eighth-round unique anagram for anonymous linking (participants are told this round does not count toward payment); (6) demographics and trait measures while the researcher is present; (7) while the researcher is out of the room, participants privately check their work using a phone to access an online Scrabble dictionary via a QR code on the productivity report, complete an anonymous productivity report for rounds 1–7, perform private self-payment from a \$14 cash envelope per the report instructions, seal the productivity report and cash envelope in a large envelope, and deposit materials in designated boxes (workbook folder and sealed large envelope separately); (8) main study debriefing sheet.

Students who participated in the pilot may also participate in the main study. Because all data are anonymous, the research team cannot monitor repeat participation without compromising anonymity.

4. Research Objectives

Our primary objective is to conduct a constructive replication of the foundational Schweitzer et al. (2004) study, correcting its methodological limitations while maintaining its core strengths. Second, we aim to investigate when and for whom performance goals lead to unethical behavior. By testing how trait loss aversion and goal proximity influence overstatement, we provide a direct empirical test of the loss aversion mechanism theorized by Ordóñez and Wu (2013). These findings will help strengthen the empirical basis of the goal-setting literature and offer new theoretical insights for behavioral ethics.

5. Research Questions or Hypotheses

H1. People with specific challenging goals (i.e., personal, mere, or reward goals) are more likely to overstate their performance than people without specific challenging goals (i.e., do-your-best condition).

H2. Challenging goal type is related to overstating performance. People with reward goals are more likely to overstate their performance than those with mere goals, who in turn are more likely to overstate their performance than those with personal goals.

H3a. People who fail to reach their goal by a smaller margin are more likely to overstate their performance than if they were to fail to reach their goal by a larger margin.

H3b. People who reach their goal are less likely to overstate their performance than if they failed to reach the goal.

H4. People with greater trait loss aversion are more likely to overstate their performance.

H5. The relationship between trait loss aversion and overstating behavior is amplified in trials where people failed their goal by a smaller margin than when they failed by a larger margin.

6. Educational Justification

This project is part of a global multi-site initiative designed to train student researchers and junior faculty in rigorous, open-science replication methodologies. By engaging with a conditionally accepted Registered Report, local co-investigators and research assistants gain firsthand experience in pre-registration, double-blind administration, and collaborative behavioral science research.

7. Recruitment Method & Script

Recruitment occurs via classroom visits using approved informational flyers. Two flyer versions are included (Appendix A for pilot sessions, Appendix A2 for main study); only the version matching the session is shown. Both pilot and main recruitment materials are supplied for HSIRB review. Instructors may offer course credit or bonus points. The main-study flyer mentions a small cash reward of at least \$7.00 with opportunity for up to \$14.00; the pilot flyer does not mention cash compensation.

Pilot script: You are invited to participate in a pilot session for a research study examining decision-making during cognitive tasks. This session takes approximately one hour, has

received IRB ethics clearance, and helps refine procedures before main data collection. There is no cash compensation. Participation is voluntary and anonymous.

Main study script: You are invited to participate in a research study examining decision-making during cognitive tasks. This study takes approximately one hour and has received IRB ethics clearance. Depending on your instructor's policy, you may receive course credit, bonus points, or a small cash reward of at least \$7.00 with opportunity for up to \$14.00 total. Participation is voluntary and anonymous.

8. Subject Inclusion & Exclusion Criteria

Inclusion:

- Undergraduate students or working professionals
- 18 years of age or older

Exclusion:

- Individuals under 18 years of age

9. Subject Benefits & Costs

Benefits to Subjects:

Pilot session participants receive no cash compensation. They may receive course credit or bonus points at their instructor's discretion and receive a written appreciation letter explaining how pilot data support calibration for the main study.

Main study participants receive direct financial compensation through the self-payment procedure, keeping at least \$7.00 and up to \$14.00 depending on their assigned instructions and performance (averaging about \$10.00). They also receive a detailed debriefing sheet explaining the logic and value of open-science replication research.

Benefits to Others/Society:

Strengthening the empirical foundation of the dark-side of goal setting literature, generating a shared understanding between goal setting and behavioral ethics scholars, and developing novel theoretical insights to guide organizational practices.

Payment/Compensation:

Pilot session: No cash compensation is provided.

Main study only: All main-study participants receive an envelope containing \$14.00 at the start of the session. Payment amounts are stated on each participant's productivity report during the private self-payment step. Those in the 'do your best', 'mere goal', and 'personal goal' conditions are instructed to keep \$10.00. Those in the 'reward goal' condition are instructed to keep \$7.00 plus an additional \$1.00 for each scored round (rounds 1–7) in which they met the site-calibrated goal (on average, approximately \$10.00). Unearned cash is returned in the cash envelope, which is sealed with the productivity report in a large envelope while the researcher is out of the room. Participant-facing materials describe a small cash reward of at least \$7.00 with opportunity for up to \$14.00 total.

Costs:

Participation requires approximately one hour of time. There are no financial costs to the subjects.

10. Basis of Exemption / Justification

The study qualifies for exempt review under Category 2 of the revised Common Rule (45 CFR 46.104(d)(2)). Data are collected via behavioral tests (an anagram task) and survey procedures. No participant names, IDs, or other identifying information are recorded on any data collection sheets, ensuring participants cannot be identified, directly or through linked identifiers. While we correlate task performance with self-reported productivity reports using a unique eighth-round anagram, this linking mechanism is entirely anonymous and never associated with any student identity.

11. Statement of Risk & Mitigation

Risk Statement:

The study presents minimal risk. The anagram task may cause mild frustration or performance anxiety, but no more than everyday schoolwork or standard puzzles. While participants have an opportunity to over-report performance or keep unearned money, all data are completely anonymous and cannot be linked to individual identities.

Mitigation:

To protect participants, the design is double-blind, and no names or student IDs are ever written on the study workbooks or productivity reports. Signed consent forms are collected and stored separately. The researcher leaves the room during grading, self-payment, and report submission. A unique eighth-round anagram is used for anonymous data linking. Finally, participants receive a detailed debriefing, and contact information for the Nicholls Counseling Center is provided.

12. Data Collection, Storage, & Access

Collection Methods:

Data collection uses paper-and-pencil materials, including a task workbook (practice and performance anagram rounds, goal commitment, trait loss aversion, and demographics), a separate productivity report for self-reported performance, and a follow-up survey for manipulation and data quality checks. Hard copies are collected in a secure box.

Storage:

Hard-copy workbooks and productivity reports are stored in a locked office in the Department of Management and Marketing at Nicholls State University. De-identified, anonymized electronic datasets are maintained on secure, password-protected computers and shared publicly on the Open Science Framework (OSF) to facilitate open-science collaboration.

Confidentiality Procedures:

Participants remain completely anonymous. No names, email addresses, or student IDs are written on the workbooks, productivity reports, or survey forms. Signed consent forms are stored in a separate locked cabinet from the raw data sheets. Anonymized data are linked via a unique eighth-round anagram rather than any personal identifier, and the de-identified spreadsheet is shared publicly on the OSF.

Access:

The principal investigator and local co-investigators have access to the raw paper materials. De-identified, anonymous data are shared with our research collaborators across the multi-site initiative and made publicly available on the Open Science Framework (OSF).

Retention:

Paper materials will be retained in a locked office for five to seven years in accordance with APA guidelines, after which they will be shredded. The anonymized electronic dataset will be preserved indefinitely on the OSF for replication and meta-analysis.

13. Consent Procedures

Two written informed consent forms are included for HSIRB review: Appendix B (pilot session) and Appendix B2 (main study). Only the form matching the session is shown to participants. Each form details the study's purpose, procedures, risks, and benefits, emphasizing that participation is entirely voluntary and that participants may withdraw at any time without penalty. The main-study form describes cash compensation; the pilot form states that no cash compensation is provided. Students who choose not to participate are offered an equivalent alternative assignment for equal course credit.

Students who participated in the pilot may also participate in the main study. Because all data are anonymous, the research team cannot monitor repeat participation without compromising anonymity.

14. CITI Training Certification

All investigators have completed CITI Program training. Dr. Christopher Castille (Faculty Researchers), Dr. Ann-Marie R. Castille (Faculty Researchers), Dr. Samantha Falgout (Faculty Researchers), Dr. Kaitlin Gravois (Faculty Researchers), Dr. Adrien Maught (Social/Behavioral RCR). Certificates are pending / to be submitted upon request.

Statement of Compliance & Ethical Assurances:

By signing below, the Primary Investigator certifies that the described project will be conducted in accordance with university guidelines, the Belmont Report, and 45 CFR 46. Any protocol modifications or adverse events must be reported immediately to the board.

Primary Investigator Signature

Date

College HSIRB Representative Signature

Date

HSIRB Chairperson Signature (Approval)

Date



NICHOLLS STATE UNIVERSITY
HUMAN SUBJECTS INSTITUTIONAL REVIEW BOARD
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Appendix A

Pilot Recruitment Flyer

A Study in Decision Making

You are invited to participate in a pilot session for a multi-site research study aimed at understanding decision-making in work settings. This study is part of a larger collaborative initiative (ARIM) and has received formal ethics clearance from the Human Subjects Institutional Review Board (HSIRB) at Nicholls State University (Protocol: IRB 2024-07-30-001 CBA). Investigators: **Dr. Christopher Castille** (Primary Investigator), **Dr. Ann-Marie R. Castille**, **Dr. Samantha Falgout**, **Dr. Kaitlin Gravois**, and **Dr. Adrien Maught** (Co-Investigators), Nicholls State University.

What will I do?

You will be asked to complete a decision-making task involving anagram puzzles (word puzzles) in a classroom or computer lab on the Nicholls State University campus. You will also complete brief questionnaires about the task, a brief personality scale, and standard demographics. Participation takes **no more than 1 hour** of your time.

Is participation confidential?

Yes, completely. All data collected in this study are anonymous. Your name and email are never associated with your answers, workbooks, or self-reported scores. Data are stored securely using de-identified code IDs.

What are the requirements?

- Must be currently enrolled as a student at Nicholls State University.
- Must be 18 years of age or older.
- Fluent in English and comfortable working on basic word problems.

Sign up for a session

Scan the QR code to choose an open time slot. **No account or login required.** You will receive a booking reference and PIN only — no name or email is collected.



Appendix A2

Main Study Recruitment Flyer

A Study in Decision Making

You are invited to participate in a multi-site research study aimed at understanding decision-making in work settings. This study is part of a larger collaborative initiative (ARIM) and has received formal ethics clearance from the Human Subjects Institutional Review Board (HSIRB) at Nicholls State University (Protocol: IRB 2024-07-30-001 CBA). Investigators: **Dr. Christopher Castille** (Primary Investigator), **Dr. Ann-Marie R. Castille**, **Dr. Samantha Falgout**, **Dr. Kaitlin Gravois**, and **Dr. Adrien Maught** (Co-Investigators), Nicholls State University.

What will I do?

You will be asked to complete a decision-making task involving anagram puzzles (word puzzles) in a classroom or computer lab on the Nicholls State University campus. You will also complete brief questionnaires about the task, a brief personality scale, and standard demographics. Participation takes **no more than 1 hour** of your time.

What will I receive?

In appreciation of your time, you will receive a small cash reward of at least **\$7.00**, with an opportunity to receive up to **\$14.00** total.

What are the requirements?

- Must be currently enrolled as a student at Nicholls State University.
- Must be 18 years of age or older.
- Fluent in English and comfortable working on basic word problems.

Is participation confidential?

Yes, completely. All data collected in this study are anonymous. Your name and email are never associated with your answers, workbooks, or self-reported scores. Data are stored securely using de-identified code IDs.

Sign up for a session

Scan the QR code to choose an open time slot. **No account or login required.** You will receive a booking reference and PIN only — no name or email is collected.



Appendix B

Pilot Informed Consent Statement



INFORMED CONSENT STATEMENT/INFORMATION LETTER

A Study in Decision Making — Pilot Session

You are invited to participate in a pilot session for a research study aimed at understanding decision-making in work settings. This pilot session helps the research team calibrate study materials and refine administration procedures before main data collection begins. The study is being conducted by Dr. Christopher Castille (Associate Professor of Management, Principal Investigator), Dr. Ann-Marie R. Castille (Associate Professor of Management, Co-Investigator), Dr. Samantha Falgout (Assistant Professor of Accounting, Co-Investigator), Dr. Kaitlin Gravois (Assistant Professor of Marketing, Co-Investigator), and Dr. Adrien Maught (Assistant Professor of Marketing, Co-Investigator) from Nicholls State University. This study is funded in part by the Nicholls State University Research Council and is part of a larger, multi-site research project coordinated through the Advancement of Replications Initiative in Management (ARIM).

Information

You are invited to participate because you are a student enrolled in a participating Business Administration course at Nicholls State University (for example, BSAD courses) or were otherwise invited to take part in this pilot session. If you agree to participate, you will attend an in-person research session in a campus laboratory. During the session, you will complete a decision-making task involving anagram puzzles (word puzzles) in a paper workbook. The purpose of this pilot session is to help us refine study procedures and establish performance benchmarks for the subsequent main study; pilot data are not used to test the study hypotheses. You will first complete brief practice rounds to learn the task, then complete several timed performance rounds. You will also complete a brief productivity report and brief questionnaires about the task, individual difference measures (including personality traits), and basic demographics. There is no cash self-payment step in the pilot session. Participation in this session will take no more than 1 hour to complete. Because understanding the full scientific purpose of the study during the session could influence responses, you will receive a brief written appreciation letter at the end of your session; a complete debriefing explaining the study's purpose and findings will be provided after data collection at Nicholls State University is complete.

Risks

There are no known serious risks associated with participation in this study. The anagram task may cause mild frustration or performance anxiety, but no more than everyday schoolwork or standard puzzles.

Benefits

The researchers cannot guarantee any direct benefits from participation. However, an important contribution of scientific research into organizational behavior is to provide organizations, managers, and employees with data so they can better understand how business and social practices influence employee behavior. In this pilot session, you will be helping the research team prepare materials and procedures for the main study.

Confidentiality of Your Information

Approximately 50 students from Nicholls State University may be invited to complete this pilot session. All data collected will be anonymous. Please do not put any identifying information (e.g., name) on your workbook, productivity report, or survey materials. Signed consent forms will be collected and stored separately from your study materials. Any information that could potentially identify participants will be immediately removed from the data. No information that would potentially identify a participant will be used in the analyses and/or results. Furthermore, because the interest of this study is in the average responses of the entire group of participants, you will not be identified individually in any way in any written reports of this research.

Any raw data will be kept for a period of at least 5 to 7 years per APA guidelines following publication of the research. Paper materials will be stored in a locked office in the Department of Management and Marketing at Nicholls State University. Electronic copies of de-identified data will be stored on secure, password-protected computers maintained by Nicholls State University. Only the researchers affiliated with Nicholls State University conducting this study will have access to the raw data.

Anonymized data related to your participation may be shared with our collaborators at other universities participating in this multi-site project. It may also be stored on the Open Science Framework (OSF), which is an online platform for data storage and management. The OSF is a secure and reputable platform widely used for research data storage and sharing, ensuring compliance with data protection and privacy standards. The data may also be submitted to a peer-reviewed journal (i.e., the data may be publicly available). The data will be completely anonymized prior to submission and will be presented in aggregate form in publications. This process is integral to the research process as it allows other researchers to verify results and avoid duplicating research. Though the data is posted for the purpose listed, other individuals may access these

data by accessing the open access repository. Although the dataset without identifiers may be shared publicly, your identity will always remain confidential. If you have concerns about anonymized data sharing, please discuss them with the researcher before agreeing to participate.

Remuneration

You will not receive cash compensation for participating in this pilot session. Depending on your instructor's policy, you may receive course credit or bonus points for participation; any such credit is offered at your instructor's discretion and is not provided by the research team in cash.

Contact

This study has been reviewed and received ethics clearance through the Human Subjects Institutional Review Board (HSIRB) at Nicholls State University (Protocol: IRB 2024-07-30-001 CBA). If you have questions for the Board regarding your rights as a research participant, contact the HSIRB Chairperson (Dr. Alaina Daigle, 985-448-4697, alaina.daigle@nicholls.edu) or your college HSIRB representative at Nicholls State University.

For all other questions, after receiving this letter, or if you would like additional information to assist you in reaching a decision about participation, please contact Dr. Christopher Castille (Principal Investigator) at christopher.castille@nicholls.edu.

Participation

Your participation in this pilot session is voluntary. You may decide not to participate or to withdraw from this session at any time by advising the researcher and may do so without penalty. If you withdraw before data collection is completed, your data will be destroyed. You have the right to skip any question(s)/procedure(s) you choose.

Once you have completed the session, you can no longer withdraw your data, as the data will not be identifiable.

Students who participated in this pilot session may also be eligible to participate in the main study. Because all data are anonymous, the research team cannot monitor whether the same individual participates in both phases without compromising anonymity.

Feedback and Publication

The results of the study may be submitted to peer-reviewed academic journals for publication consideration and/or be presented at a conference. If you would like a copy of the results, please contact Dr. Christopher Castille (Principal Investigator) at christopher.castille@nicholls.edu.

CONSENT

By signing this consent form, you are not waiving your legal rights or releasing the investigator(s) or involved institution(s) from their legal and professional responsibilities.

By signing this form, you agree that you have had the opportunity to ask any questions and receive satisfactory answers.

I agree to participate in this pilot session.

Name of Participant

Date

Appendix B2

Main Study Informed Consent Statement



INFORMED CONSENT STATEMENT/INFORMATION LETTER

A Study in Decision Making — Main Study

You are invited to participate in a research study aimed at understanding decision-making in work settings. The study is being conducted by Dr. Christopher Castille (Associate Professor of Management, Principal Investigator), Dr. Ann-Marie R. Castille (Associate Professor of Management, Co-Investigator), Dr. Samantha Falgout (Assistant Professor of Accounting, Co-Investigator), Dr. Kaitlin Gravois (Assistant Professor of Marketing, Co-Investigator), and Dr. Adrien Maught (Assistant Professor of Marketing, Co-Investigator) from Nicholls State University. This study is funded in part by the Nicholls State University Research Council and is part of a larger, multi-site research project coordinated through the Advancement of Replications Initiative in Management (ARIM).

Information

You are invited to participate because you are a student enrolled in a participating Business Administration course at Nicholls State University (for example, BSAD courses) or were otherwise invited to take part in this study. If you agree to participate, you will attend an in-person research session in a campus classroom or laboratory. During the session, you will complete a decision-making task involving anagram puzzles (word puzzles) in a paper workbook. The purpose of this task is to help us understand how people approach decision-making tasks in work settings. You will first complete brief practice rounds to learn the task, then complete several timed performance rounds under instructions assigned to you at random. You will also be asked to complete brief questionnaires about the task, individual difference measures (including personality traits), and basic demographics. After the timed rounds, while the researcher is out of the room, you will privately check your work and complete an anonymous productivity report (you may use your phone only to access an online Scrabble dictionary via a QR code on that report). Participation in this study will take no more than 1 hour to complete. Because understanding the full scientific purpose of the study during the session could influence responses, you will receive a brief written appreciation letter at the end of your session; a complete debriefing explaining the study's purpose and findings will be provided after data collection at Nicholls State University is complete.

Risks

There are no known serious risks associated with participation in this study. The anagram task may cause mild frustration or performance anxiety, but no more than everyday schoolwork or standard puzzles.

Benefits

The researchers cannot guarantee any direct benefits from participation. However, an important contribution of scientific research into organizational behavior is to provide organizations, managers, and employees with data so they can better understand how business and social practices influence employee behavior. In this study, you will be helping us understand the factors that determine employees' decision making and how managers can influence employees' behavior.

Confidentiality of Your Information

Approximately 200 students from Nicholls State University will be invited to complete this study. All data collected will be anonymous. Please do not put any identifying information (e.g., name) on your workbook, productivity report, or survey materials. Signed consent forms will be collected and stored separately from your study materials. Any information that could potentially identify participants will be immediately removed from the data. No information that would potentially identify a participant will be used in the analyses and/or results. Furthermore, because the interest of this study is in the average responses of the entire group of participants, you will not be identified individually in any way in any written reports of this research.

Any raw data will be kept for a period of at least 5 to 7 years per APA guidelines following publication of the research. Paper materials will be stored in a locked office in the Department of Management and Marketing at Nicholls State University. Electronic copies of de-identified data will be stored on secure, password-protected computers maintained by Nicholls State University. Only the researchers affiliated with Nicholls State University conducting this study will have access to the raw data.

Anonymized data related to your participation will be shared with our collaborators at other universities participating in this multi-site project. It will also be stored on the Open Science Framework (OSF), which is an online platform for data storage and management. The OSF is a secure and reputable platform widely used for research data storage and sharing, ensuring compliance with data protection and privacy standards. The data may also be submitted to a peer-reviewed journal (i.e., the data may be publicly available). The data will be completely anonymized prior to submission and will be presented in aggregate form in publications. This process is integral to the research process as it allows other researchers to verify results and avoid

duplicating research. Though the data is posted for the purpose listed, other individuals may access these data by accessing the open access repository. Although the dataset without identifiers may be shared publicly, your identity will always remain confidential. If you have concerns about anonymized data sharing, please discuss them with the researcher before agreeing to participate.

Remuneration

In appreciation of your participation, you will receive a small cash reward of at least \$7.00, with an opportunity to receive up to \$14.00 total depending on the instructions assigned to you during the session and your performance on the anagram task.

At your workstation you will find a sealed cash envelope containing \$14.00 in one-dollar bills. After you complete the timed anagram rounds and brief questionnaires, the researcher will leave the room so you can work privately. You will check your work, complete an anonymous productivity report for the seven scored rounds (round eight does not count toward payment), and take from the cash envelope only the amount specified on that report for your assigned instructions. Place any unearned cash back in the cash envelope, seal the cash envelope and productivity report together in the large envelope provided, and deposit your materials as directed. The research team cannot verify your self-reported performance or link the amount you keep to your identity.

Because payment details depend on your randomly assigned instructions, the specific amount you may keep is stated on your productivity report during the session, not on this consent form.

Contact

This study has been reviewed and received ethics clearance through the Human Subjects Institutional Review Board (HSIRB) at Nicholls State University (Protocol: IRB 2024-07-30-001 CBA). If you have questions for the Board regarding your rights as a research participant, contact the HSIRB Chairperson (Dr. Alaina Daigle, 985-448-4697, alaina.daigle@nicholls.edu) or your college HSIRB representative at Nicholls State University.

For all other questions, after receiving this letter, or if you would like additional information to assist you in reaching a decision about participation, please contact Dr. Christopher Castille (Principal Investigator) at christopher.castille@nicholls.edu.

Participation

Your participation in this study is voluntary. You may decide not to participate or to withdraw from this study at any time by advising the researcher and may do so without penalty. If you withdraw before the private self-payment step, you will receive \$7.00 for your time and your data will be destroyed. If you withdraw after that step, you may keep any cash you already took per the productivity report instructions, but your data will be destroyed. You have the right to skip any question(s)/procedure(s) you choose.

Once you have completed the study, you can no longer withdraw your data, as the data will not be identifiable.

Students who participated in the pilot session may also participate in the main study. Because all data are anonymous, the research team cannot monitor whether the same individual participates in both phases without compromising anonymity.

Feedback and Publication

The results of the study may be submitted to peer-reviewed academic journals for publication consideration and/or be presented at a conference. If you would like a copy of the results, please contact Dr. Christopher Castille (Principal Investigator) at christopher.castille@nicholls.edu.

CONSENT

By signing this consent form, you are not waiving your legal rights or releasing the investigator(s) or involved institution(s) from their legal and professional responsibilities.

By signing this form, you agree that you have had the opportunity to ask any questions and receive satisfactory answers.

I agree to participate in this study.

Name of Participant

Date

Appendix C

Anagram Task Workbook (Study Instrument)

A Study in Decision Making

Please do not open this folder until you have been instructed to do so.

WORKBOOK

Practice Round

You will be given one minute to create as many words as you can.

Rules:

1. Each word must be at least two letters long.
2. No proper nouns (e.g., Larry, Mexico, Boston Red Sox).
3. Use each of the 7 letters only once per word.
4. Words can only be used in one form (e.g., “task” or “tasks”; not “task” and “tasks”).

Trial Round # 1

UONHMEY

[illegible]

Trial Round # 2

AODJGIP

[illegible]

GOAL:

DO YOUR BEST to create as many valid English words as you can within one minute using the letters provided.

Please answer the following questions about your goal:

	Not at all	Slightly	Moderately	Quite a bit	Extremely
1. How committed are you to this goal?	1	2	3	4	5
2. To what extent do you care about this goal?	1	2	3	4	5
3. How dedicated are you to this goal?	1	2	3	4	5
4. To what extent have you chosen to be committed to this goal?	1	2	3	4	5

After answering the questions above, please wait. The researcher will tell you when to continue.

Round #1

AEDBKUG

[illegible]

Round #2

OELBJAM

[illegible]

Round #3

UADQWER

[illegible]

Round #4

EASCKIY

[illegible]

Round #5

OADMHUP

[illegible]

Round #6

OELHMAZ

[illegible]

Round #7

OASFKEV

[illegible]

Round #8

EXPERIMENTAL ANAGRAM

RAVOLET

[illegible]

DEMOGRAPHICS

You are free to skip any questions that you wish. Your responses to these questions will not be used to identify you. We only ask them to study demographic trends.

1. What is your age? _____
2. What is your gender?
 - A. Woman
 - B. Man
 - C. I identify my gender as (please specify): _____
3. What is your study level?
 - A. Undergraduate (e.g., Bachelor's) Degree
 - B. Graduate (e.g., Master's) Degree
 - C. Other: _____
4. What is your major? _____
5. Are you currently enrolled in a university program that is taught primarily in English?
 - A. Yes
 - B. No
6. Is English your first language?
 - A. Yes
 - B. No
7. If English is not your first language, how many years have you spoken English? _____

8. Please select the response option that describes your proficiency in written English.

A. Proficient	Understands almost everything effortlessly, summarize complex information from multiple sources, and communicates themselves with precision, fluency, even in nuanced or complex situations.
B. Advanced	Understands demanding texts and implicit meanings, express themselves fluently with minimal effort, and use language effectively in social, academic, and professional settings, producing clear, well-structured texts on wide subjects.
C. Upper Intermediate	Understands main ideas of complex texts and interacts fluently, producing clear, detailed texts on various topics.
D. Intermediate	Understands key points of familiar topics and handles common situations, expressing simple ideas and personal opinions.
E. Pre Intermediate	Understands basic phrases and communicates in routine tasks on familiar topics with simple descriptions.
F. Beginner/Elementary	Understands and uses basic phrases for everyday needs, communicating simply if the other person speaks slowly.

9. Ethnic origin refers to the ethnic or cultural origins of one's ancestors. An ancestor is usually more distant than a grandparent. Please select the option(s) that best describe your ethnic origin:

- A. North American origins (Indigenous, First Nations, Inuit, Metis, and others)
- B. European origins
- C. Caribbean origins
- D. Latin, Central and/or South American origins
- E. African origins
- F. Asian origins
- G. Oceanian origins
- H. Prefer to specify _____

FOLLOW-UP SURVEY

Please indicate how much you agree with each of the following statements **in general**, not right now or in this moment.

	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
1. When making a decision, I think much more about what might be lost than what might be gained.	1	2	3	4	5
2. The pain of losing money matters more than the pleasure of gaining the same amount of money.	1	2	3	4	5
3. I feel nervous when I have to make a decision that may lead to loss.	1	2	3	4	5
4. The pain from losing something matters much more to me than the pleasure from getting it.	1	2	3	4	5
5. Experiencing a major loss stays in my mind longer than experiencing a major gain.	1	2	3	4	5
6. A potential failure scares me more than a potential success encourages me.	1	2	3	4	5
7. The suffering that comes with losses can be fully offset by the pleasure that comes from gains.	1	2	3	4	5

Appendix D

Participant Productivity Report (Main Study)

PRODUCTIVITY REPORT

**PLEASE DO NOT SIGN YOUR NAME OR PROVIDE ANY IDENTIFYING
INFORMATION ON THIS PRODUCTIVITY REPORT.**

Revisit your workbook and cross out any words in your workbook that are invalid.

Again, the rules for determining whether a word is ‘valid’ are:

1. Each word must be at least two letters long.
2. No proper nouns (e.g., Larry, Mexico, Boston Red Sox).
3. Use each of the 7 letters only once per word.
4. Words can only be used in one form (e.g., “task” or “tasks”; not “task” and “tasks”).

Please cross out any words you created that are invalid (i.e., break the rules).

You may use the online Scrabble dictionary available via the QR code below:



PRODUCTIVITY REPORT

The purpose of this worksheet is for you to report your productivity relative to a specific number of words. Please do the following:

Below, place an X into the box indicating whether or not you created the target number of words. For example, if you created 6 words for the first anagram, then you place an “X” into the box “Created 6 or fewer words”.

ANAGRAM	Created 6 or fewer words	Created 7 or more words
1. AEDBKUG		
2. OELBJAM		
3. UADQWER		
4. EASCKIY		
5. OADMHUP		
6. OELHMAZ		
7. OASFKEV		
8. RAVOLET		

Report continued on the next page.

Please complete these two final questions about your participation in this study:

1. Please read the four options below. Then, select the one option that best describes your approach to the anagram task. *In each round of the anagram task...*

A.	My goal was to “ do my best ”.
B.	My goal was to create a specific number of words. I was told that “ in previous studies, many students were able to create this number of words ”.
C.	My goal was to create a specific number of words. I was told that I would earn money for each round in which I met the goal .
D.	My goal was to create a specific number of words. I was not told anything about students in previous studies nor about receiving money for each round in which I met the goal.

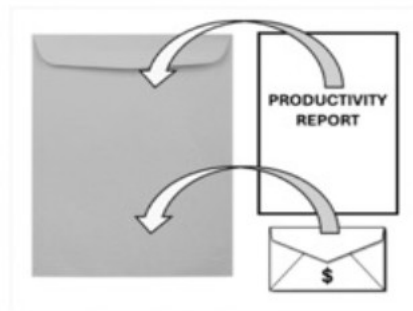
2. In your honest opinion, should we use your data from this study? *You will receive your compensation regardless of how you answer.*

- A. Yes
B. No

If you selected “No”, please briefly explain *why* we should not use your data.

Report continued on the next page.

**After completing this report, take
\$10.00 from the cash envelope.
This is for you to keep.**



**Place the envelope containing the remaining cash and this productivity report into the large envelope.
Seal the envelope.**

Follow the directions on the slide to deposit your materials.

**Once you have deposited your materials, the study is
complete and you may leave.**

Appendix E

Pilot Study Appreciation Letter



FEEDBACK/APPRECIATION LETTER

A Study in Decision Making

Dr. Christopher Castille, Associate Professor of Management, Nicholls State University, christopher.castille@nicholls.edu

Dr. Ann-Marie R. Castille, Associate Professor of Management, Nicholls State University, ann-marie.castille@nicholls.edu

Dr. Samantha Falgout, Assistant Professor of Accounting, Nicholls State University, samantha.falgout@nicholls.edu

Dr. Kaitlin Gravois, Assistant Professor of Marketing, Nicholls State University, kaitlin.gravois@nicholls.edu

Dr. Adrien Maught, Assistant Professor of Marketing, Nicholls State University, adrien.maught@nicholls.edu

Thank You for Your Participation

Thank you for taking part in our pilot study. Your time and effort help us prepare for our upcoming research. This letter explains the purpose of this pilot session, how your responses will be used, and what we are investigating.

Study Purpose and Calibration

The primary purpose of this session was to pilot our research materials and practice administration protocols. Specifically, we are measuring baseline student performance on the anagram task to set an empirical 90th percentile target goal for the subsequent main study. This calibration ensures that the goals we assign in the main study are challenging but realistic for our student population.

There are no experimental manipulations, hidden conditions, or partial disclosures in this pilot study. All participants were asked to do their best to create as many words as possible.

How Your Responses Were Used

During the study, you completed anagram rounds in a workbook. All data collected in this pilot study are completely anonymous. Your name and email are never associated with your answers, workbooks, or self-reported scores. Data are stored securely using de-identified code IDs.

What We Are Investigating

This pilot study is a crucial preparatory step for a larger, multi-site research project coordinated through the Advancement of Replications Initiative in Management (ARIM). By practicing our administration protocols and calibrating our performance targets, we ensure the high quality and reliability of the data we will collect in the main study.

Questions or Concerns

If you have questions about the study or these procedures, please contact Dr. Christopher Castille (Principal Investigator) at christopher.castille@nicholls.edu.

This study has been reviewed and received ethics clearance through the Human Subjects Institutional Review Board (HSIRB) at Nicholls State University (Protocol: IRB 2024-07-30-001 CBA). If you have questions about your rights as a research participant, contact the HSIRB Chairperson, Dr. Alaina Daigle (985-448-4697, alaina.daigle@nicholls.edu), or your college HSIRB representative at Nicholls State University.

Summary of Findings

When data collection and analysis are complete, we will use the baseline performance data to set the targets for the main study. If you would like a summary of the aggregated findings from the overall project, please contact Dr. Christopher Castille at christopher.castille@nicholls.edu.

Thank you again for your assistance. We sincerely hope you found the session worthwhile.

Appendix E2

Main Study Debriefing & Appreciation Letter



DEBRIEFING PROTOCOL / APPRECIATION LETTER

A Study in Decision Making

Dr. Christopher Castille, Associate Professor of Management, Nicholls State University, christopher.castille@nicholls.edu

Dr. Ann-Marie R. Castille, Associate Professor of Management, Nicholls State University, ann-marie.castille@nicholls.edu

Dr. Samantha Falgout, Assistant Professor of Accounting, Nicholls State University, samantha.falgout@nicholls.edu

Dr. Kaitlin Gravois, Assistant Professor of Marketing, Nicholls State University, kaitlin.gravois@nicholls.edu

Dr. Adrien Maught, Assistant Professor of Marketing, Nicholls State University, adrien.maught@nicholls.edu

Thank You for Your Participation

Thank you for taking part in our study. Your time and effort help us understand how people complete decision-making tasks in work settings. This letter explains the full purpose of the study, what you were told earlier, and what we are investigating.

Study Purpose and Disclosure

At the start of the session, we told you that the purpose of the study was to examine how people approach decision-making tasks involving word puzzles. That description was only partly complete. The full purpose of this study is to examine whether the type of performance instructions you received determines the degree to which people overstate their performance on these word puzzles (that is, report solving more anagrams than they actually solved).

We did not tell you the full purpose at the beginning because knowing it in advance could influence how you approached the task and how you reported your performance. Withholding this information until now is a standard and approved research practice called partial disclosure. We are telling you the complete purpose now so you understand what the study was designed to test.

How Your Responses Were Used

During the study, you completed anagram rounds in one booklet and reported your performance in a separate productivity report. In the final performance round, you solved a unique anagram (a letter combination assigned only to you). This allowed us to match your actual workbook performance with your self-reported performance without collecting your name or any other identifying information.

While you were working, the researcher left the room so you could check your work privately using your phone to access an online Scrabble dictionary via a QR code on your productivity report. You also determined how much cash to keep from the envelope on your desk, based on the instructions on your productivity report. Any unearned cash was returned anonymously in the large envelope with your productivity report. We recorded how much money participants kept, but this information cannot be linked to your identity.

Conditions in This Study

Participants were randomly assigned to one of four instruction conditions:

"Do your best" — You were instructed to do your best to create as many words as you can.

"Mere goal" — You were given a specific performance target (the site-calibrated goal from the pilot session; for example, to create 7 or more words) and told that in previous studies many students were able to achieve this target.

"Personal goal" — You were given the same specific performance target (for example, to create 7 or more words) but without the comparison to previous studies.

"Reward goal" — You received the same target as the mere goal condition and were also told that your cash payment depended on whether you met the target in each round.

You also completed brief measures of commitment to the assigned target and individual differences related to how unpleasant it is to fall short of a target (sometimes called loss aversion).

What We Are Investigating

This study is part of a larger, multi-site research project coordinated through the Advancement of Replications Initiative in Management (ARIM). We are replicating and extending a foundational study on how specific, challenging performance targets can influence both actual performance and self-reported performance. We are especially interested in whether effects depend on people's commitment to their assigned target and on individual differences in loss aversion.

Questions or Concerns

If you have questions about the study or these procedures, please contact Dr. Christopher Castille (Principal Investigator) at christopher.castille@nicholls.edu.

This study has been reviewed and received ethics clearance through the Human Subjects Institutional Review Board (HSIRB) at Nicholls State University (Protocol: IRB 2024-07-30-001 CBA). If you have questions about your rights as a research participant, contact the HSIRB Chairperson, Dr. Alaina Daigle (985-448-4697, alaina.daigle@nicholls.edu), or your college HSIRB representative at Nicholls State University.

Summary of Findings

When data collection at Nicholls State University is complete, we will analyze results across all participating sites. If you would like a summary of the aggregated findings, please contact Dr. Christopher Castille at christopher.castille@nicholls.edu.

Thank you again for your assistance. We sincerely hope you found the session worthwhile.

Appendix F

Approved UWaterloo Master Protocol

#46813 - A Study in Decision Making

Protocol Information

Review Type	Status	Submission Number	Approval Date
Expedited	Approved	3	Mar 20, 2026
Renewal Date	Expiration Date	Initial Approval Date	Initial Review Type
Sep 29, 2026	Oct 22, 2026	Oct 21, 2024	Expedited

Feedback

Approval Comment

This amendment has received ethics clearance. If you have any questions please contact Karen Pieters at karen.pieters@uwaterloo.ca.

Protocol Amendment Form

Amendment

Describe the proposed changes and why these changes are being made. Refer to the [guideline for modifications](#) to find review tips for preparing an amendment request. Review the full application and ensure that you:

- 1) Revise/edit all sections that relate to the amendment.
- 2) Upload all revised or new study materials to the appropriate section. Use track changes or highlight within the document to show where changes have been made.

This study is part of a large multi-site replication initiative. Since our preliminary data collection (STUDY 1) last year, this study was submitted to Psychological Science in the form of a registered report. As part of the review process, we made several changes to the protocol. We have now received approval from Psychological Science (in the form of a conditional acceptance) to go ahead and collect data for all conditions. We refer to the planned data collection as "STUDY 2" below.

General Information

Only the Principal Investigator/Faculty Supervisor can submit the application. This acts as a signature indicating approval of the application.

Principal Investigator / Faculty Supervisor

Annika Hillebrandt

Department

Psychology

Study title

A Study in Decision Making

General Questionnaire

Indicate the type of application you would like to complete
Standard application

Please confirm:

I understand that the type of applications listed above determine the form I am about to complete. If I have chosen the incorrect form I acknowledge that I may need to complete a new application.

People

Research team members based at the University of Waterloo.

Ensure all information in this table is completed.

Person

Annika Hillebrandt

Waterloo Department

Psychology

Email Address

ahillebr@uwaterloo.ca

Phone

Researcher Role

Principal Investigator

Permissions

Full Access

Mandatory Training

All University of Waterloo undergraduate and graduate students, faculty, and staff must complete the [TCPS 2 2022 CORE tutorial](#) prior to submitting an application for review. See [instructions](#) or email researchethics@uwaterloo.ca if you have questions.

REQUIRED: Upload a copy of the TCPS 2 2022 certificate or a screen shot showing module completion. Applications where the certificate/screen shot is not uploaded will be sent back to the researcher and not reviewed.

[TCPS2 2022 - Hillebrandt.pdf](#)

Ensure all information in this table is completed.

Person

Caitlin Belfiore

Waterloo Department

Psychology

Email Address

cbelfior@uwaterloo.ca

Phone

Researcher Role

Co-Investigator

Permissions

Full Access

Mandatory Training

All University of Waterloo undergraduate and graduate students, faculty, and staff must complete the [TCPS 2 2022 CORE tutorial](#) prior to submitting an application for review. See [instructions](#) or email researchethics@uwaterloo.ca if you have questions.

REQUIRED: Upload a copy of the TCPS 2 2022 certificate or a screen shot showing module completion. Applications where the certificate/screen shot is not uploaded will be sent back to the researcher and not reviewed.

[TCPS2 2022 - Belfiore.pdf](#)

Ensure all information in this table is completed.

Person

Megan Herrewynen

Waterloo Department

Psychology

Email Address

mrhverre@uwaterloo.ca

Phone

Researcher Role

Co-Investigator

Permissions

Full Access

Mandatory Training

All University of Waterloo undergraduate and graduate students, faculty, and staff must complete the [TCPS 2 2022 CORE tutorial](#) prior to submitting an application for review. See [instructions](#) or email researchethics@uwaterloo.ca if you have questions.

REQUIRED: Upload a copy of the TCPS 2 2022 certificate or a screen shot showing module completion. Applications where the certificate/screen shot is not uploaded will be sent back to the researcher and not reviewed.

[TCPS 2 Core Certificate Megan Herrewynen.pdf](#)

Do you have any investigators external to the University of Waterloo?

Yes

People external to the University of Waterloo

First Name

Laurie

Last Name

Barclay

Position Title

Professor and Lang Chair in Leadership

Institution/organization

University of Guelph - Gordon S. Lang School of Business and Economics

Email Address

lbarcl01@uoguelph.ca

Researcher role

Co-Investigator

First Name

Jenny

Last Name

Wang

Position Title

PhD Candidate

Institution/organization

University of Guelph - Gordon S. Lang School of Business and Economics

Email Address

jwang76@uoguelph.ca

Researcher role

Student investigator

First Name

Christine

Last Name

Hwang

Position Title

PhD Student

Institution/organization

University of Guelph - Gordon S. Lang School of Business and Economics

Email Address

hwangc@uoguelph.ca

Researcher role

Student investigator

Are you the lead PI for this multi-jurisdictional project (e.g., you are named the lead PI for the funding)?

Yes

If this study involves external researchers from a Canadian institution eligible to receive and administer Tri-Agency funds, are you requesting a single ethics review with your Canadian collaborating institution(s), facilitated by Waterloo's Research Ethics office staff?

Before responding to this questions please review Waterloo's guideline on [multi-jurisdictional research and research ethics review](#).

Note: Single ethics review is only possible for studies deemed minimal risk by the reviewing REB.

No and I am aware that my collaborating investigators may still need to obtain an ethics review through their institutional REB.

General details

Is this new study related to any previous application?

No

What is the estimated start and end date for the study?

Start Date (Date you anticipate beginning the recruitment of participants)

October 14, 2024

End Date (Date you expect the study will end, for example, the date when there will be no further contact with participants or when the data analysis will be completed for US funded research)

September 30, 2026

Does this research require approval from a UWaterloo departmental committee?

Not a department requirement

What is the level of the research to be conducted? Choose one.

Faculty, tenured, tenure track

Will this study involve Wilfrid Laurier University or Local hospitals covered by the Waterloo Wellington Research Ethics Board (WWREB) (established by Cambridge Memorial Hospital, Grand River Hospital and St. Mary's General Hospital)?

No

Has a version of this study been disapproved or rejected by any Research Ethics Board/Committee?

No

Study description

State your research question(s).

STUDY 1: This pilot study will help us determine realistic performance goals for our main study (which we are planning to conduct in Winter 2025). Participants will be given a series of anagrams and asked to create as many words as possible across multiple rounds. STUDY 2: This study uses the same task as Study 1 (which has now been completed) to determine to what extent, when, and for whom goals promote unethical behavior.

Provide a clear, detailed description of the purpose, hypothesis, aim, and objectives of this study.

STUDY 1: We are conducting a pilot study to determine realistic performance goals for our main study. Participants will be given a series of anagrams and asked to create as many words as possible across multiple rounds. We will be using these data to determine how many anagrams participants can be expected to complete in a given amount of time.

STUDY 2: Study 2 is an extension of Study 1. The ultimate goal of the study is to address key methodological limitations of the original study by Schweitzer et al. (2004), test theoretically important boundary conditions (e.g., trait loss aversion, goal proximity), and enhance generalizability by contributing to a multi-site global project. We hypothesize the following:

H1. People with specific challenging goals (i.e., personal, mere, or reward goals) are more likely to overstate their performance than people without specific challenging goals (i.e., do-your-best condition). H2. Challenging goal type is related to overstating performance. People with reward goals are more likely to overstate their performance than those with mere goals, who in turn are more likely to overstate their performance than those with personal goals. H3a. People who fail to reach their goal by a smaller margin are more likely to overstate their performance than if they were to fail to reach their goal by a larger margin. H3b. People who reach their goal are less likely to overstate their performance than if they failed to reach the goal. H4. People with greater trait loss aversion are more likely to overstate their performance. H5. The relationship between trait loss aversion and overstating behavior is amplified in trials where people failed their goal by a smaller margin than when they failed by a larger margin.

Provide background information, a rationale, and justification for conducting this study. Describe why the research is being done and what research has already been done in this area. Be sure to explain why this research is important.

Our ultimate goal is to conduct a replication of Schweitzer, Ordoñez, and Douma (2004) – a highly-influential publication in the management discipline. Replication studies are important to ensure the replicability and generalizability of scientific findings. Schweitzer et al. studied the effects of goal-setting on students' performance and over-reporting on an anagram task. STUDY 1: For the pilot study, we are using the same anagram task as Schweitzer et al. We will be using the data from the pilot study to determine how many anagrams students can complete (on average) and determine realistic performance goals for the main study (STUDY 2). STUDY 2: Our first objective is to conduct a global replication of Schweitzer et al.'s (2004) foundational study. We use a “constructive” replication approach that addresses limitations of the original study while retaining its key strengths (see Köhler & Cortina, 2021). Our second objective is to advance our foundational understanding related to what extent, when, and for whom goals prompt unethical behavior. By examining the effects of trait loss aversion and goal proximity on unethical behavior, we provide a direct empirical test of the previously theorized role of loss aversion (Ordoñez & Wu, 2013). Answering these questions is important for strengthening the empirical foundation of the dark-side of goal-setting literature, generating a shared understanding between goal-setting and behavioral ethics scholars, and developing novel theoretical insights that can guide future research.

In a maximum of 250 words, provide a non-scientific lay language description that summarizes the project outlining the purpose, anticipated benefits, and basic procedures. Write this summary as if it would be read by members of the general public who are not familiar with academic terms or acronyms. Use language suitable for a media release.

STUDY 1: We are conducting a pilot study to determine realistic performance goals for our main study. Participants will be given a series of anagrams and asked to create as many words as possible across multiple rounds. We will be using these data to determine how many anagrams participants can be expected to complete in a given amount of time. They will also complete a survey measuring their attitudes towards the task and basic demographic information. This study will inform the procedures of a future study, which will ultimately contribute to the academic literature by advancing our understanding of the effects of goal-setting on individuals' behavior. STUDY 2: The goal setting literature currently recognizes that setting specific, challenging goals can prompt people to overstate their performance. We are interested in exploring this effect further by examining to what extent, when, and for whom goals promote this effect. To do so, we are replicating a highly cited paper along with other sites. In the experiment, participants will be assigned different goals. They will be asked solve a series of anagrams and report their score, which is designed to allow us to determine who has over-reported their performance. This research will contribute to the academic literature by advancing our understanding of the effects of goal-setting on individuals' behavior.

What is the study design?

STUDY 1: N/A (a pilot study that will inform the procedures of a subsequent study; single condition). STUDY 2: An experiment. Participants will be randomly assigned to one of the following conditions: 1) "do your best," 2) "mere goal," 3) "reward goal," or 4) "personal goal."

Is this a pilot study?

No

Sample Size

What is the expected sample size? Outline the number of participants anticipated to take part in the study.

STUDY 1: 70-80. STUDY 2: 200.

Was a formal sample size calculation completed?

See [Statistics Help Desk](#) for support in calculating sample sizes.

No

Provide a rationale for the number of participants specified

STUDY 1: As this is part of a replication project, our goal is to follow Schweitzer et al. (2004)'s procedure as closely as possible. Their original pilot study included 70 participants. STUDY 2: This data collection is part of a large multi-site replication study. All sites have agreed to contribute 200 participants to the overall dataset.

Study sites

Where is this study taking place?

On a University of Waterloo campus

Are there any permissions required to conduct this study on campus?

No

Funding

Is the study funded/will it be funded?

Yes

Funding

List all funding sources that are new or ongoing

--

Funding status

Ongoing funding

Funding source is

Tri-agency / Canadian Government sponsor

Industry partner/sponsor name

Canadian Government agency

SSHRC - Social Sciences and Humanities Research Council

Charitable foundation

Other Charitable foundation name

Internal UWaterloo funding source

US federal funds

Other US federal funds sponsor

Other funding source

Program name if applicable

Insight Grant

Work-order or award number, if known

50656-10196 500

What is the expected period of funding

What is the period of funding

Funding from

April 1, 2023

Funding to

March 31, 2027

Conflict of interest

Are there any potential, perceived, or actual financial or non-financial conflicts of interest of the research team in undertaking the proposed research?

No

Benefits

Are there direct benefits of the proposed research to the study participants?

No

What are the scientific and/or scholarly benefits of the proposed research?

This study will contribute to the academic literature by advancing our understanding of the effects of goal-setting on individuals' behavior.

Participants

Participant general categories

University of Waterloo undergraduate and/or graduate students

Are you conducting research in classes with students as your participants to evaluate a teaching method or object?

No

Describe the sample in detail and list any specific inclusion/exclusion criteria for the study

Participants must be students registered with the SONA system. No other inclusion or exclusion criteria will be used.

If you are excluding people on certain characteristics provide a justification for the exclusion.

N/A

Will a screening process be used to determine eligibility in the study based on the inclusion and/or exclusion criteria identified above?

No

Recruitment

Identify from where/what sources potential participants will be recruited. If your participants are part of an organization/group please see [guidance for conducting research when participants are part of an organization or group](#) and incorporate this information into the methods section.

Database or Participant Pool

Indicate the Database or Participant Pool that will be used in the study.

SONA participant pool at the University of Waterloo.

What recruitment materials will be used? See [sample recruitment materials](#).

Other

Describe the other recruitment materials

A recruitment ad on the SONA website.

Upload your recruitment materials

Upload your recruitment materials

[Recruitment_version2_20260312.pdf](#)

Study group

Will potential participants be recruited through pre-existing relationships with members of the research team (e.g., employees, students, or patients of research team, acquaintances, own children or family members, colleagues, etc.)?

No

Methods and procedures

Which of the following will be conducted for this study?

Surveys/questionnaires

How will the survey(s) or questionnaire(s) be administered?

In-person - Paper and pen/pencil

Will quotations be used in the write-up of the study

No

For each of the procedures indicated above, provided a detailed, sequential description of how they will be used in the study. Provide one or more paragraphs describing how you are recruiting participants, obtaining consent, what participants are asked to do, and how the research team will be using the collected data.

STUDY 1: 1. Participants enter a computer lab in groups of approximately 20 students. 2. They are asked to read a consent form and given the opportunity to ask questions before signing it. 3. Participants receive a package of study materials (hard copies only) containing the anagram task. The goal of this task is to create as many words as possible from a given series of letters. 4. Participants complete 2 practice rounds of the anagram task (1 minute per round). 5. Next, they are instructed to do their best on the main task and complete a goal commitment measure. 6. Participants complete 8 more anagram trials. 7. Participants check their work. They are instructed that they can use an online Scrabble dictionary (e.g., <https://scrabble.merriam.com/>) if they wish. The Scrabble dictionary website can be accessed using the lab computer in front of them. No online/computer data will be collected/stored. 8. Participants indicate how many words they were able to create in each round (i.e., <9 or >9). 9. Finally, they complete some demographic measures and measures of their perceptions/attitudes of the anagram task. 10. Participants receive a feedback/appreciation form.

STUDY 2: 1. Participants enter a computer lab in groups of approximately 20 students (minimum of 5 students per session). 2. They are asked to read a consent form and given the opportunity to ask questions before signing it. The consent forms are then collected. 3. Participants receive a package of study materials (hard copies only) containing the anagram task. The task involves creating words from a given series of letters. 4. Participants complete 2 practice rounds of the anagram task. 5. Participants receive instructions for the main rounds of the anagram task. The specific instructions/"goal" that participants are given varies across conditions. There are four conditions to which participants are randomly assigned: "do your best," "mere goal," "personal goal," or "reward goal". In the "do your best" condition, participants are told to "do your best to create as many words as you can" on the following trials. In the "mere goal" condition, participants are given a specific challenging goal. The instructions are: "It is important that you commit to a specific yet attainable goal. Your goal is to create 8 words during the allotted 1 minute using these 7 letters. This goal is difficult, but realistic. In

previous studies, many students were able to create 8 or more words per round using the same groups of letters you are about to see." The "personal goal" condition uses the same instructions as the "mere goal" condition but omits the sentence containing the social comparison information ("In previous studies..."). This condition has been added to examine whether social comparison information may have operated as a potential confound in the original study. In the "reward goal" condition, participants receive the same instructions as the mere goal condition but are also told they will earn \$1 for each round in which they meet the goal. Following these instructions, they complete a measure of commitment to the goal. 6. Participants complete a measure of trait loss aversion and demographics. 7. The researcher(s) leave the room while participants check their work. They are instructed that they can use an online Scrabble dictionary (e.g., <https://scrabble.merriam.com/>) if they wish. The Scrabble dictionary website can be accessed using the lab computer in front of them. No online/computer data will be collected/stored. 8. Participants use their "productivity report" to indicate how many words they were able to create in each round (i.e., <7 or >6), and complete a manipulation check and data quality check. 9. Participants pay themselves by taking money out of an envelope on their desk (see Remuneration) and deposit any remaining money along with their study materials in a box. 10. Upon exiting the room, participants receive a feedback/appreciation form from one of the researchers.

Please upload any study materials related to the procedure(s)

Study material

[Workbook_version2_20260312.pdf](#)

Study material

[ProductivityReport_version2_20260312.pdf](#)

Does the study involve the administration or use of an approved drug or natural health product?

No

Will you be collecting any biological specimens?

No

Will you be creating or contributing to a bio-bank, bio-repository, registry, as part of the study?

No

Will you be doing any genetic testing or analysis?

No

Incidental and secondary findings

See [Guideline for reporting incidental and secondary findings to study participants](#)

Are any of the methods or procedures used likely (i.e., a real possibility and probability) to reveal an incidental finding (i.e., discoveries made in the course of research but that are outside the scope of the research and/or results that are outside the original purpose for which a test or procedure was conducted)?

No

Are any of the methods or procedures used likely to reveal a secondary finding (i.e., findings that are not the primary target of the test or procedure; rather, it is an additional result that is actively sought)?

No

Equipment use

Will there be any equipment used as part of this study?

No

Deception

Does the study involve deception or partial disclosure?

Yes - Partial disclosure

What is the study title that will appear on the Information-Consent Letter?

A Study in Decision Making

Fully describe the partial disclosure. Describe the information that will be withheld from the participants.

STUDY 1: Study 1 does not involve any partial disclosure or deception.

STUDY 2: Participants are told that the purpose of the study is to examine how people approach word problems. However, the full purpose of the study is to examine whether the type of goal that a person receives (i.e., a "do your best" goal, personal goal, mere goal, or reward goal; see Methods and Procedures) determines the degree to which they overstate their performance on these word problems (i.e., report solving more anagrams than they actually solved). During the study, participants will complete an anagram task in one booklet and report their performance in a separate booklet. However, the final anagram (e.g., "BNMOETR") will be different for each participant. This will allow us to match their actual performance on the anagram task (completed in one booklet) to their self-reported performance (from the other booklet). We will use this to determine whether/how much each participant over-reported their performance. Consistent with the original study by Schweitzer et al., participants will remunerate themselves by taking money out of an envelope placed on their desk (see Remuneration). While there is a possibility that participants may take more money than they should, Schweitzer et al. found this to be a very low base-rate phenomenon (only one participant in their study took unearned money). Nonetheless, we will record how much money each participant took from the envelope. This information will not be linked to the participant's identity.

Describe how participants will be debriefed about the study.

Participants will receive a feedback/appreciation letter at the end of the study session (see Feedback and Appreciation).

Upload Debriefing materials

Risks and safeguards

Considering each method or procedure to be used in this study, indicate if participants might experience any of the following risks or harms

No known or anticipated risks

Outline the criteria for stopping the study early due to safety concerns/other issues.

None

Privacy

Will demographic and/or background information be asked of participants? If so, ensure that the demographic questions have been uploaded in the methods section.

Yes

What demographic/background information will be collected?

Age

Gender

Ethnicity

Education

Other

Describe what "other" demographic/background information will be collected.

whether English is their first language, years they have spoken English, English proficiency, study major, level of study

Will demographic/background information be collected separately from names and other identifying information?

Yes

Participant identification - If applicable, include how participants will be referenced in study results.

STUDY 1: Participants will only put their SONA IDs on the study materials. These materials will not be linked to their consent forms (or other identifying information). STUDY 2: Participants will not be identified in any way. They will NOT be asked to put their name, SONA ID, or any other identifying information on any data collection materials. (They do put their name on the consent form, but the consent form cannot be linked to the other study materials in any way).

If applicable how will the key/list that links participants' codes with their actual name and/or consent forms be stored and protected? Also, outline how long the key/list will be stored.

N/A

Are there any limitations to the promise of confidentiality?

No

Will any study data be leaving the University of Waterloo, the province, or country (e.g., member of research team is located in another institution, province, or country, etc.)?

Yes

Will any identifiable participant information be leaving the University of Waterloo, the province, or country (e.g., member of research team is located in another institution, province, or country, etc.)?

Yes

Where will the study data be sent? Why is it necessary for it to leave the University of Waterloo?

We are collaborating with researchers from the University of Guelph (listed on this protocol). The raw data collected at UW may be shared with our Guelph collaborators, but will not be shared with anyone else. Our study is part of a larger, multi-site research project. Anonymous data that cannot in any way be linked to individual participants will be shared with the other research teams that are part of this multi-site research project to contribute to a single, large dataset. The data will also be posted on the Open Science Framework (OSF), a secure online platform.

Explain what data will be leaving the University of Waterloo, who will receive it, why they need access, and what safeguards will be used to protect the identity of participants and the privacy of their data.

The raw data (i.e., hard copies) may be shared with our collaborators at Guelph (listed on this protocol) for the purpose of data analysis and verification. The data will not contain any information that could potentially identify a participant. Once they have been entered into electronic format (e.g., spreadsheet), the anonymous will be shared with the other teams of our multi-site research project. They will also be posted on the Open Science Framework (OSF), allowing the researchers to check and compare data across sites.

Describe the measures in place to ensure secure transfer of study data outside of the University.

No identifying information will be collected. Our collaborators at the University of Guelph will have access to the raw data. The anonymous data, which cannot be linked to the participants in any way, may be shared with our collaborators at other sites via university email or the OSF.

Has a research data agreement or data transfer agreement been created?
Yes

Will any collected data or information be entered into a database for future use?
No

Are there other members of the research team who are not named on this application (e.g., co-op students, research assistants, or other temporary personnel) who may carry out specific tasks involved in your study?

No

Will individual participant identities be confidential in the publication or release of the study findings?

Yes

Data storage

What type(s) of data will be collected for this study?

Hard copy / Paper records

Electronic files

For each type of information collected, identify where the data will be stored

Hard copies will be stored in a locked office at the University of Waterloo.

The researchers on this protocol may also store electronic copies of the anonymous data on password-protected computers, secure internal databases or encrypted external drives. Electronic files (containing anonymized data only) will also be stored on the Open Science Framework (OSF) for data storage and management. The OSF is a secure and reputable platform widely used for research data storage and sharing, ensuring compliance with data protection and privacy standards.

For each type of data collected, identify the minimum retention period

7 years

Data Management

Are there plans to link the data collected with other data sets, databases, or registries?

No

The [Tri-Agency Open Access Policy on Publications](#) and some journals are requesting that research data be provided to an open access repository to promote the availability of findings, to enhance transparency and share with the widest possible audience.

Do researchers plan to make the data-set available in an online repository/archive?

Yes

Identify the repository or database and outline whether the dataset will be anonymized or if it will include identifiable information. If this is not yet known please state this.

We will be using the Open Science Framework (OSF) for data storage and management. The OSF is a secure and reputable platform widely used for research data storage and sharing, ensuring compliance with data protection and privacy standards. The data will NEVER contain identifiable information (i.e., participants' names, SONA IDs, etc. will never be associated with any of the materials). Only anonymous data will be submitted to the OSF.

Do you have a data management plan? See the Library [Research Data Management: Get Started guide](#).

Yes

Consent and Withdrawal

What member(s) of the research team will be responsible for obtaining informed consent?

Caitlin Belfiore, Megan Herrewynen, Jenny Wang, Christine Hwang

Is there a relationship between the potential participant(s) and the person obtaining consent?

No

How will consent be obtained

Written signature on a consent form

Upload Information and Consent Materials - See [resources](#) and [samples](#) for creating information consent letters. Refer to the guide for creating an information consent letter on this [webpage](#).

Upload Information and Consent Materials

[ConsentForm_version2_20260312.pdf](#)

Study group

All participants

Do you anticipate that you will need to make special accommodations for your participant group?

No

Do you anticipate needing to put in place any special procedures when obtaining informed consent?

No

Will consent need to be re-documented throughout the life of this study?

No

Describe the participants rights and processes for withdrawing consent.

If participants withdraw from the study before data collection is complete, their data will be destroyed. Participant will still receive remuneration.

Outline what will be done with the participant's information (data, samples, etc.) if they withdraw from the study.

Data will be destroyed.

Will any individuals taking part in this study be unable to provide their own informed consent?

No

Remuneration

Will there be remuneration provided to show appreciation for a participant's time, effort, skills, etc. to take part in the study?

Yes

Type of remuneration

Cash

Participation credits (*Psychology only)

Dollar amount:

STUDY 2: Renumeration will vary depending on condition. All participants will receive an envelope containing \$14.00. Participants in the "do your best," "mere goal," and "personal goal" conditions will instructed to keep \$10.00. Participants in the "reward goal" condition will be instructed to keep \$7.00 plus an additional \$1.00 for each round in which they met their goal (on average, we expect they will receive approximately \$10.00). Thus, all participants in the study will receive at least \$7.00.

Explain the participation credits

STUDY 1: 1 SONA credit for participating in a 1-hour study.

If a participant withdraws from the study will remuneration be pro-rated?

No, participants will receive maximum remuneration

Will participants incur any expenses by participating in the study?

No

Feedback and Appreciation

How will you show appreciation to participants for taking part in the study?

Participants will receive a written feedback/appreciation letter.

When will feedback/appreciation be provided to participants (e.g., immediately after the session, at the end of a survey, mail results at time X.)?

Immediately after the session.

Upload Feedback/Appreciation materials

Upload Feedback/Appreciation materials

[Feedback_version2_20260312.pdf](#)

Study group

All participants

How can participants learn about the study results/obtain a summary of the findings if interested?

Participants are invited to visit the PI's website: <https://uwaterloo.ca/justice-affect-and-morality-lab/>

Other Details

Provide any other information relevant to this study you wish to explain to the Research Ethics Board reviewers or to the staff in the Office of Research Ethics.

N/A

Other Attachments

Upload any additional study documents.

Attachments

Attestation

As the Principal Investigator/Faculty Supervisor/Local Investigator, I attest to the following:

- I will ensure all co-investigators, collaborators, and student investigators listed on this application have reviewed the application contents and will conduct the study according to the application/protocol.
- I am aware that any changes made to the research must be reviewed and provided clearance before the changes are implemented. Change requests (i.e., an amendment) are to be submitted through the system. I am also aware ethics clearance for this study is valid for only 12 months unless I renew the study prior to the ethics clearance expiry date. If an annual renewal report is NOT submitted through the system prior to the expiry date, the study will be suspended, all work on the study must stop, and Research Finance will be notified which will result in a hold being put on the funds associated with this study.
- I agree to comply with the [Tri-Council Policy Statement \(TCPS2\)](#) for conducting research with human participants and with University of Waterloo policies and guidelines when conducting this study (e.g., [statement on human participant research](#), [IST policies](#), etc.).
- I confirm I have read the [University of Waterloo Research Integrity guidelines](#) and I agree to comply with the policies and guidelines of my profession or discipline regarding the ethical conduct of research involving humans.
- Proper safeguards as to confidentiality and security of data will be maintained and data will be securely stored in accordance with University of Waterloo standards and applicable privacy legislation.

By submitting this application I agree to the above attestations and will ensure the research is conducted accordingly

Only the Principal Investigator/Faculty Supervisor can submit the application. This acts as a signature indicating approval of the application.

This is the end of the application form. Click submit in the right menu if you are ready to send it to the Research Ethics Office.

Appendix G

Psychological Science Registered Report Manuscript

Abstract

Specific challenging goals can improve motivation and performance, but they can also risk promoting unethical behavior. While this effect is widely recognized, goal-setting scholars and behavioral ethics scholars have debated *to what extent, when, and for whom* goals promote unethical behavior. To address these questions, we conducted a pre-registered, multi-site constructive replication of Schweitzer et al. (2004) – the first study to systematically demonstrate that specific challenging goals can increase unethical behavior. Our replication addresses key methodological limitations of the original study (e.g., by disentangling mere goals from social comparisons), tests theoretically important boundary conditions (e.g., trait loss aversion, goal proximity), and enhances generalizability by leveraging research sites from across the globe. In doing so, our replication aims to strengthen the empirical foundation of the dark-side of goal-setting literature, promote a shared understanding between goal-setting and behavioral ethics scholars, and contribute novel theoretical insights that can guide future research.

Registered Report: Revisiting the Effect of Specific Challenging Goals on Unethical Behavior: Insights from a Global, Preregistered, Constructive Replication Study

Goal Setting Theory (Locke & Latham, 1990) is widely considered one of the most practically useful and scientifically valid theories in the social sciences (Miner, 2003; Mitchell & Daniels, 2003). Countless studies support the theory's central proposition that goals can increase attention, effort, and persistence, thereby enhancing performance (Locke & Latham, 2019, 2020). However, goals need to be specific and challenging to have these motivating properties. Goals that lack specificity (e.g., "do your best") or are easy to attain tend to be relatively ineffective (Latham & Locke, 2006; Locke & Latham, 1990, 2015).

Despite their beneficial effects on motivation and performance, the use of specific challenging goals has not been without controversy. Behavioral ethics scholars have warned that specific challenging goals can lead people to resort to unethical behavior when they are unable to meet the goal, even if there is no external reward associated with achieving the goal (for a review, see Ordóñez & Welsh, 2015). Some have even concluded that specific challenging goals are like "a prescription-strength medication" that should be used sparingly due to its "powerful and predictable side effects" (Ordóñez et al., 2009, p. 6).

Tracing this argument to its roots, Schweitzer et al. (2004) were the first to demonstrate that specific challenging goals can prompt unethical behavior. Participants in their study were asked to solve a series of anagrams under time pressure – a task widely used by goal-setting scholars. Results indicated that participants who had been randomly assigned to receive a specific challenging goal ("create nine or more words") were more likely to overstate their performance than those who had been instructed to do their best. Importantly, this unethical behavior not only occurred when participants were promised a financial reward for attaining

specific challenging goals (“reward goal” condition), but also when they were given a specific challenging goal *without* a financial reward (“mere goal” condition). This suggests that goals are sufficient to elicit unethical behavior, even in the absence of financial incentives.

Schweitzer et al.’s (2004) paper has been highly influential, garnering over 1,100 citations and inspiring a research program on the “dark side” of goal-setting (see Ordóñez & Welsh, 2015; Welsh et al., 2019). However, goal-setting scholars and behavioral ethics scholars have debated the degree to which individuals are likely to engage in unethical behavior in pursuit of specific challenging goals (e.g., Latham & Itzhakov, 2024). While both groups of scholars agree that people sometimes act unethically to achieve goals, critical questions remain regarding the foundational findings reported by Schweitzer et al., including *to what extent*, *when*, and *for whom* these effects are likely to occur. Answering these questions is important for strengthening the empirical foundation of the dark-side of goal-setting literature, generating a shared understanding between goal-setting and behavioral ethics scholars, and developing novel theoretical insights that can guide future research.

Our first objective is to conduct a global replication of Schweitzer et al.’s (2004) foundational study. We use a “constructive” replication approach that addresses critical limitations of the original study while retaining its key strengths (see Köhler & Cortina, 2021). We identified potential limitations that may have threatened the original study’s internal validity based on the published paper as well as consulted with a proponent of the dark side of goal-setting (Lisa Ordóñez, who was also a co-author of the Schweitzer et al. paper) and with one of the original authors of Goal Setting Theory and critic of the dark side approach (Gary Latham). Table 1 summarizes the identified limitations and our improvements. For example, a key limitation relates to the presence of social comparisons in the original “mere goal” condition,

which raises the question of whether social comparisons rather than mere goals were driving the original effects. To address this, the replication adds a “personal goal” condition that disentangles mere goals from social comparisons to enhance precision and eliminate this potential confound (i.e., ensure that “mere goals” are indeed sufficient for prompting unethical behavior).

Our second objective is to advance our foundational understanding related to what extent, when, and for whom goals prompt unethical behavior. Drawing on Prospect Theory (Kahneman & Tversky, 1979), scholars have argued that goals can serve as reference points, such that people tend to experience falling short of a goal as a loss (Heath et al., 1999). Furthermore, because people are generally loss averse, they may be motivated to engage in unethical behavior to reach an unmet goal (e.g., by overstating their performance; Ordóñez & Wu, 2013). However, while it follows from Prospect Theory that falling short of a goal is generally aversive, individuals differ in trait loss aversion, such that some experience losses as more aversive than others (Li et al., 2021). Integrating these theoretical insights, we propose that trait loss aversion should be positively associated with unethical behavior. Due to Prospect Theory’s principle of diminishing sensitivity (see Heath et al., 1999), we also hypothesize that the effect of trait loss aversion on unethical behavior is particularly strong when individuals miss their goal by a small (versus large) margin. Table 2 provides an overview of our hypotheses and research plan.

By examining the effects of trait loss aversion and goal proximity on unethical behavior, we provide a direct empirical test of the previously theorized role of loss aversion in explaining unethical behavior (Ordóñez & Wu, 2013). Additionally, given that trait loss aversion is an individual difference that is sensitive to cultural influences (Wang et al., 2017), examining this trait at each site of our global replication study may also provide insight into whether the hypothesized effects are sensitive to cultural variations.

Table 1. *Methodological Improvements Made in Our Constructive Replication of Schweitzer et al. (2004).*

Design/analysis choice by Schweitzer et al.	Limitation/validity threat	How we address it
1. A student sample ($N = 154$) from a US university.	Generalizability (external validity) and low statistical power (statistical validity) are concerns.	A large sample of students will be recruited from a number of universities across the globe.
2. Only “reward goal” participants had an opportunity for theft (i.e., were asked to return unearned money whereas those in other conditions were provided with the full payment).	The original method did not allow theft in the conditions without financial rewards to be measured (Latham & Itzhakov, 2024). Providing envelopes of cash to all participants ensures that the methodology reduces potential demand effects (which can create internal validity concerns) while ensuring that all participants have the opportunity to steal.	All participants will receive an envelope of cash. This gives all participants an opportunity to steal, regardless of their condition. We ensure that participants are randomly assigned to conditions with experimenter-blind protocols during data collection to reduce the potential for bias.
3. The instructions used in the “mere goal” condition may have invoked social comparisons.	Schweitzer et al. (2004) argued that specific challenging goals are sufficient to prompt unethical behavior. However, student participants in the “mere goal” condition were told that “in previous studies many students were able to [meet the goal]” (p. 425). These instructions may have invoked social comparisons and motivated people to cheat (e.g., to accomplish an interpersonal goal; Van Yperen et al., 2011; Welsh et al., 2019). Thus, specific challenging goals were confounded with social comparison. We consider this an internal validity threat.	We include a new condition – the “personal goal” condition – in which no reference is made to other people’s performance.
4. Used a measure of goal commitment with low reliability.	Schweitzer et al. explored the relationships between goal commitment and unethical behavior as well as task performance. However, the goal commitment measure they used (adapted from Hollenbeck et al., 1989) has known reliability problems (Klein et al., 2013). This is a concern for construct validity.	We include a more recent version of the goal commitment measure that was designed to be more reliable (Klein et al., 2013).

Table 2. *Comprehensive Summary of Research Plan.*

Research Question	Hypothesis	Simulation Parameters (Betas in log-odds)	Effect Size Justification	Power: 100% Effect Size	Power: 75% Effect Size	Power: 50% Effect Size	Effect Tested	Interpretation of Potential Results
Are people with specific challenging goals more likely to overstate their performance than people without specific challenging goals?	H1. People with specific challenging goals (i.e., personal, mere, or reward goals) are more likely to overstate their performance than people without specific challenging goals (i.e., do-your-best condition).	Overstatement rates by group: dyb = 0.03 mere = 0.08 reward = 0.13 personal = 0.055 Regression estimates: dyb = -3.48 mere = 1.03 reward = 1.58 personal = 0.63	Incidence reported by Schweitzer et al. (2004).	Mere: ~100% Reward: ~100% Personal: ~99.7%	Mere: ~100% Reward: ~100% Personal: ~95.4%	Mere: ~99.4% Reward: ~100% Personal: ~62.7%	Model includes: <i>mere_goal</i> , <i>reward_goal</i> , <i>personal_goal</i> Test <i>mere_goal</i> , <i>reward_goal</i> , and <i>personal_goal</i> (dummy-coded, using “Do-Your-Best” as the reference group). Interpret effects individually.	If H1 is supported, this indicates that people with specific challenging goals are more likely to overstate their performance than those without such goals.
Are different types of specific challenging goals associated with different degrees of overstating behavior?	H2. Challenging goal type is related to overstating performance. People with reward goals are more likely to overstate their performance than those with mere goals, who in turn are more likely to overstate their performance than those with personal goals.	Overstatement rates by group: dyb = 0.03 mere = 0.08 reward = 0.13 personal = 0.055 Regression estimates: dyb = -3.48 mere = 0.53 reward = 0.50 personal = 0.63	Incidence reported by Schweitzer et al. (2004).	Mere vs Reward: ~100% Personal vs Mere: ~97.6%	Mere vs Reward: ~99.8% Personal vs Mere: ~81.3%	Mere vs Reward: ~94.6% Personal vs Mere: ~48.6%	Model includes: <i>mere,vs,reward</i> , <i>personal,vs,mere</i> , <i>dyb.vs.mere</i> Test <i>mere,vs,reward</i> & <i>personal,vs,mere</i> (custom, progressive dummy codes: lower triangular 1s, remaining 0s).	If both contrasts are significant and in the predicted direction, people with reward goals are more likely to overstate their performance than those with mere goals; both are more likely to overstate their performance than

Research Question	Hypothesis	Simulation Parameters (Betas in log-odds)	Effect Size Justification	Power: 100% Effect Size	Power: 75% Effect Size	Power: 50% Effect Size	Effect Tested	Interpretation of Potential Results
							Interpret effects individually.	those with personal goals.
Does goal proximity affect overstating behavior?	H3a. People who fail to reach their goal by a smaller margin are more likely to overstate their performance than if they were to fail to reach their goal by a larger margin.	$proximity_w = 1.79$	Extrapolated from Schweitzer et al. (2004): overall rate = .379 → near goal rate = .337 → $OR = 6$ → log-odds = 1.79	~100%	~100%	~100%	Model adds: $proximity_w, proximity_b, goal_met_w, goal_met_b$ Test $proximity_w$ predictor (continuous)	A significant, positive effect indicates that the closer a person is to achieving their goal, the more likely they are to overstate their performance.
Does goal attainment affect overstating behavior?	H3b. People who reach their goal are less likely to overstate their performance than if they failed to reach the goal.	$goal_met_w = -5.29$	Motivation to overstate should be negligible when the goal is met. A 99.5% decrease in overstatement: $pp = 0.005$ → log-odds = -5.29	~100%	~100%	~100%	Model adds: $proximity_w, proximity_b, goal_met_w, goal_met_b$ Test $goal_met_w$ predictor (binary)	A significant, negative effect indicates that when a person meets their goal, they are less likely to overstate their performance than if they did not meet their goal.

Research Question	Hypothesis	Simulation Parameters (Betas in log-odds)	Effect Size Justification	Power: 100% Effect Size	Power: 75% Effect Size	Power: 50% Effect Size	Effect Tested	Interpretation of Potential Results
Are differences in overstating between individuals explained by individual differences in loss aversion?	H4. People with greater trait loss aversion are more likely to overstate their performance.	$\lambda_z = 0.55$	Garbarino et al. (2019): $r = 0.15 \rightarrow$ $OR = 1.73 \rightarrow$ log-odds = 0.55	~100%	~99.4%	~58.3%	Model adds: λ_z Test λ_z predictor (continuous, grand-mean standardized).	A significant, positive effect indicates that higher trait loss aversion is associated with a greater likelihood of overstating.
Is the relationship between trait loss aversion and overstating behavior affected by goal proximity?	H5. The relationship between trait loss aversion and overstating behavior is amplified in trials where people failed their goal by a smaller margin than when they failed by a larger margin.	$proximity_w \times \lambda_z = 0.30$	35% increase in odds of λ_z for each point closer to goal than individual's average, assuming $\lambda_z = 0.55$	~99.5%	~98.6%	~90.4%	Model adds: $\lambda_z \times (proximity_w proximity_b goal_met_w goal_met_b)$ Test $\lambda_z \times proximity_w$ interaction	A significant, positive effect indicates that loss aversion's effect on overstating is amplified when people are nearer (<i>versus</i> farther) to reaching their goal.

Method

Participants and Sites

Data will be collected at 8 independent sites across the globe. All contributors have agreed to sample university students who are enrolled in English-taught programs. Each site has committed to gathering data for 200 participants across the four conditions of the main study (see Analyses section for details). Consistent with Schweitzer et al. (2004), all data will be collected in-person using a traditional pen-and-paper approach.¹

Procedure

Goal calibration. Consistent with Schweitzer et al.'s (2004) procedure, each site will calibrate a “specific challenging goal” (i.e., 90th percentile performance goal) that will be used in the main study. Participants (approximately $N = 70$ per site) are given a set of 7 letters and instructed to create as many words as possible within 1 minute. They perform this task 9 times (2 practice trials, 7 main trials) with different combinations of letters. The results from the 7 main trials are used to calculate the 90th percentile of performance on the anagram task at each site, which will be used in the main study.

Main study conditions. Participants will be randomly assigned to four conditions: “do your best,” “mere goal,” “personal goal,” or “reward goal”. In the “do your best” condition, participants are told to “*do your best to create as many words as you can*” on the following trials.

¹ In addition to the 8 primary sites that will follow the exact procedure outlined in the Procedure section, several additional sites will slightly deviate from this procedure to provide data for supplementary analyses. These supplementary sites do not have the financial means to provide monetary rewards to participants and as such cannot include a “reward goal” condition nor provide envelopes with money. However, they will closely follow all other procedures and collect data for the remaining three conditions (i.e., mere goal, personal goal, and do-your-best). Participants at these sites will receive non-monetary compensation for their participation (e.g., course bonus credit). Including these supplementary sites is considered important for the diversity of our initiative (i.e., it ensures that less affluent sites can also contribute data). Given that the supplementary sites will not collect data for the reward goal condition, they will not be included in our main analyses/hypothesis tests. However, we will conduct supplementary analyses to test if the findings pertaining to the non-reward goal conditions are consistent between our primary sites and the supplemental ones.

In the “mere goal” condition, participants are given a specific challenging goal (i.e., the 90th percentile identified in the goal calibration study). The instructions are:²

It is important that you commit to a specific yet attainable goal. Your goal is to create ____ words during the allotted 1 minute using these 7 letters. This goal is difficult, but realistic. In previous studies, many students were able to create ____ or more words per round using the same groups of letters you are about to see.

The “personal goal” condition uses the same instructions as the “mere goal” condition but omits the sentence containing the social comparison information (“In previous studies...”). This condition has been added to examine whether social comparison information may have operated as a potential confound in the original study. In the “reward goal” condition, participants receive the same instructions as the mere goal condition but are also told they will earn \$1 for each round in which they meet the goal.

In Schweitzer et al.’s (2004) study, all participants were given cash at the start of the study. Participants in the “do your best” and “mere goal” conditions were given \$10 and told to keep the full amount. Participants in the “reward goal” condition were given \$14 and told to keep \$2 for each round in which they met the goal; they returned any unearned money along with their productivity reports in sealed envelopes. We deviate from Schweitzer et al. by giving all participants – regardless of condition – an envelope with \$14. Participants in the “do your best,” “mere goal,” and “personal goal” conditions are instructed to keep \$10. Participants in the “reward goal” condition are instructed to keep \$7 plus \$1 for each round in which they met the goal.³ This deviation has three benefits. First, giving participants more money than is required for

² The blanks are to be filled by the goals established from each site’s calibration study.

³ For non-US sites, we identified a pay rate similar to that used by Schweitzer et al. (2004) using a tool (https://docs.google.com/spreadsheets/d/1sCyeTfRXBOATFYIViqQc_Z2XmqkIKxmDx_hemW46nTc/edit?gid=536822520#gid=536822520) built upon OECD’s purchasing power parities and local currency specifics.

payment allows unethical behavior beyond overstating (i.e., theft) to be examined in all conditions, a recommendation made in the literature (Latham & Itzhakov, 2024). Second, giving all participants identical envelopes with money allows for random assignment to conditions and ensures that the experimenters conducting the study remain blind to the conditions. Third, giving participants in the “reward goal” condition a base amount (\$7) ensures that all participants are fairly compensated for participating in the study, even if they do not reach the goal.

Main study procedure. To provide a sense of anonymity, we aim to schedule participants in groups of 20, with a minimum of 5 participants per session. Participants will also verbally be assured that their data will remain anonymous.

Each participant will receive a folder containing (a) a “workbook” that includes a measure of trait loss aversion, the anagram task along with condition-specific instructions, manipulation checks, and a demographic survey, and (b) a “productivity report” for self-reporting their performance on the anagram task. The workbook and productivity report for the “do your best condition” can be found in the OSF repository: https://osf.io/f5u39/?view_only=e457ca87096f4de9a5fa7ae16cc839e7. The materials for the other conditions are identical except for the condition-specific instructions described above.

Participants first complete two practice rounds of the anagram task, with participants being instructed (via their workbook and verbally by the experimenter) to “create as many words as you can” in one minute. Following these practice rounds, they read their condition-specific instructions, complete our measure of goal commitment, and complete a 7-round anagram task (1

Rewards at non-US sites aim to translate US sites’ rewards into comparable terms, not only in the total amount participants may receive but also by providing equal opportunities for theft. Since cash envelopes in the US contain 14 one-US dollar banknotes, we translated the reward into 14 local banknotes/coins while ensuring that the total amount was as close as possible to that obtained by directly translating into local currencies using purchasing power parity.

minute per round). They are also given an 8th anagram and told that its purpose is to pilot-test anagrams for future studies. Unbeknownst to participants, this round contains an anagram that is unique to each participant, thus allowing their data to be linked to their productivity report (described below). Next, participants complete a measure of trait loss aversion and provide demographic information.

Following the anagram task, participants are instructed to check their own work. The experimenters then leave the room while participants check their work using an online Scrabble dictionary and complete a “productivity report”, which involves checking one of two boxes indicating whether they reached their goal for each round of the anagram task. Participants will be allowed at least 25 minutes to complete this stage. Finally, participants complete a manipulation check and data quality check. To assure participants that their goal attainment claims cannot be verified, they will deposit their anagram workbooks (which contain their created words) and productivity reports (which contain their claims regarding goal attainment) in separate boxes.

Measures

Unethical behavior. We will use Schweitzer et al.’s (2004) method to assess unethical behavior, operationalized as whether or not participants overstated performance on the task.

Like Schweitzer et al. (2004), we will also measure theft (i.e., participants taking unearned money from their envelope). However, given that Schweitzer et al. did not formally hypothesize an effect on theft and found theft to be a very low base-rate phenomenon (i.e., only one participant took unearned money), we only include this variable in an exploratory manner.

Proximity to goal attainment. For each round, we will calculate the difference between number of correct words and the site-specific 90th percentile goal as a measure of proximity to

goal attainment.

Trait loss aversion will be measured with Li et al.'s (2021) scale (7 items; e.g., "When making a decision, I think much more about what might be lost than what might be gained").

The response scale ranges from (1) *strongly disagree* to (5) *strongly agree*.

Consistent with Schweitzer et al. (2004), we will include a measure of *goal commitment* in an exploratory manner. Goal commitment will be measured with Klein et al.'s (2013) scale (4 items; e.g., "How committed are you to this goal?"). The response scale ranges from (1) *strongly disagree* to (5) *strongly agree*.

Demographics will include age, gender, level of study, study major, English language proficiency, and ethnicity.⁴

Manipulation check. The prompt for the manipulation check will be: "Please read the four options below. Then, select the one option that best describes your approach to the anagram task. In each round of the anagram task...", followed by 4 items: "My goal was to "do my best," "My goal was to create a specific number of words. I was told that in previous studies, many students were able to create this number of words," "My goal was to create a specific number of words. I was told that I would earn money for each round in which I met the goal," "My goal was to create a specific number of words. I was not told anything about students in previous studies nor about receiving money for each round in which I met the goal."

Data quality check. Participants will be asked whether we should use their data (*yes/no*) and to provide a brief explanation if they believe it should not be used. Participants will be assured they will receive their compensation regardless of how they respond.

Analyses

Proposed Statistical Models

⁴ Due to regulatory restrictions in some countries, it is not possible to collect ethnicity data for all sites.

We have three important differences in our study that necessitate a different analytical strategy than the one employed by Schweitzer et al. (2004). First, we examine trait loss aversion and proximity to goal attainment as potential moderators. Second, we measure overstating behavior at the trial level as opposed to aggregating trials. This approach allows specific predictors (e.g., goal proximity) to be associated with the odds of overstating on a particular trial (as opposed to overstating in general). Third, to account for multiple data collection sites, we have introduced a nesting structure that was not present in the original study. Specifically, we have trial level variables (predictors and outcome) at level 1 and individual/participant level variables at level 2. With nested data, a multilevel model (MLM), also known as a hierarchical linear model or mixed model, is most appropriate. Because our outcome variable is binary (0 = did not overstate performance; 1 = overstated performance), a logistic MLM approach (i.e., generalized linear mixed models; GLMM) is necessary to account for the non-normal distribution of the model residuals. To aid interpretation of the MLMs, results will be reported using odds ratios and their 95% confidence intervals.

Due to concerns around adequate power for testing moderation effects at any particular site, we decided against a meta-analytic approach, opting instead to pool the data across all sites into a single dataset for analysis. With a relatively small number of sites overall ($k = 8$) and no specific hypotheses at the site level, we decided to include site as a fixed-effects covariate in our models. This decision is further justified by the reality that cluster specific interpretations and estimation can be difficult with discrete outcomes especially when combined with a small number of clusters overall (see McNeish et al., 2017). By including site as a fixed effects covariate, we are able to control for and quantify differences due to site, despite having no specific hypotheses about these differences.

Our analysis will involve a hierarchical approach, where we incrementally add predictors and interaction terms to our regression model. This approach ensures that we estimate coefficients that are a close match to the condition differences in Schweitzer et al. (2004) in earlier models, while also demonstrating how these effects may change as a result of adding in the other theoretically relevant predictors of goal proximity and loss aversion. Our statistical models will be binomial (i.e., logit-link) two-level GLMMs with random intercepts. We will use the `glmer()` function with the “bobyqa” optimizer from the `lme4` package (Bates et al., 2015) in R (R Core Team, 2024) to estimate the statistical models. The first logit-model to be tested is the unconditional model, only including fixed and random intercepts (by participant) to examine the unadjusted intraclass correlation coefficient (ICC), an estimate of the variance accounted for by random intercepts. Small ICCs (closer to 0) suggest that one person’s overstatement in a trial is no more closely associated with their other trials than to those of the other participants. Large ICCs (closer to 1) suggest that nearly all variation in the dependent variable is due to interindividual differences.

Model 1 will test the group differences in Hypotheses 1, controlling for site effects.

$$\begin{aligned} \text{logit}(\text{Overstate}_{ij}) = & \gamma_{00} + \gamma_{01} \text{mere.goal}_j + \gamma_{02} \text{reward.goal}_j \\ & + \gamma_{03} \text{personal.goal}_j + \gamma_{03} : \gamma_{09} \text{site}_j + u_{0j} \end{aligned}$$

Here, i denotes a particular trial within person, j , γ_{00} is the intercept, representing the mean log-odds for the “do your best” condition, and *mere.goal*, *reward.goal*, and *personal.goal* are all dummy coded variables, representing the change in log-odds relative to the reference group, “do your best”. Differences between sites are controlled for via *site_j* variables, and u_{0j} is the

participant-specific random intercept.

The subsequent statistical models will build on previous equation hierarchically, adjusting the group contrasts from dummy-codes to successive difference contrast codes.

$$\begin{aligned} \text{logit}(\text{Overstate}_{ij}) = & \gamma_{00} + \gamma_{01} \text{DYB} . \text{vs} . \text{personal}_j + \gamma_{02} \text{personal} . \text{vs} . \text{mere}_j \\ & + \gamma_{03} \text{mere} . \text{vs} . \text{reward}_j + \gamma_{10} \text{proximity} . w_{ij} + \gamma_{20} \text{goal} . \text{met} . w_{ij} + \\ & \gamma_{01} \text{proximity} . b_j + \gamma_{02} \text{goal} . \text{met} . b_{ij} + \gamma_{03} \text{lambda} a_j + \\ & \gamma_{13} (\text{proximity} . w_{ij} \cdot \text{lambda} a_j) + \gamma_{23} (\text{goal} . \text{met} . w_{ij} \cdot \text{lambda} a_j) + \\ & \gamma_{04} (\text{proximity} . b_j : \text{lambda} a_j) + \gamma_{05} (\text{goal} . \text{met} . b_j : \text{lambda} a_j) + \gamma_{06} : \gamma_{013} \text{site} e_j + u_{0j} \end{aligned}$$

As with Model 1, i represents a trial within a person, j , γ_{00} is the intercept, representing the grand mean log-odds after controlling for the other variables in the model, $\text{site} e_j$ are site effects, and u_{0j} is the random intercept.

The other four models will address Hypotheses 2-5. Model 2 will include *DYB.vs.personal*, *personal.vs.mere*, and *mere.vs.reward* in addition to the intercepts and site effects. These variables use successive difference contrast coding, which progressively compares groups pairs from the weakest (do your best) to strongest (reward) goal condition. Model 3 will add proximity to and achievement of site goals (*proximity* & *goal.met*; explained below) as predictors. Model 4 will add *lambda*: trait loss aversion. Finally, Model 5 will include the interactions between *proximity* and *goal.met* with loss aversion.

The *proximity* and *goal.met* variables are unique among the predictors. Though we predict that motivation to overstate performance increases as goal proximity decreases, the effect should be reversed if people meet the goal. Simply, participants should not have reason to overstate if they were successful. This threshold-like effect necessitates the variable being

divided in two: *proximity*, the continuous difference between actual performance and the site-specific goal, and *goal.met*, a binary indicator of whether the trial's goal was met. As these variables are measured at the trial level, it is also important to parse the within-subjects and between-subjects variance associated with them. First, our model will include *proximity.w*, the continuous, within-subjects variable representing proximity from attaining the goal in a given trial. Second, the model will include *proximity.b*, the individual's average proximity score across all trials, which will ensure that we separate the between person effects from our proximity variable. To facilitate interpreting the moderation in Hypothesis 5, *proximity.b* will be grand-mean centered. Third, *goal.met.w* is the within subjects variable representing when a goal on a trial is met. Fourth, *goal.met.b* represents the grand-mean centered rate of goal attainment for each person. Of these four, *proximity.w* and *goal.met.w* are the variables related to our hypotheses, yet person means *proximity.b* and *goal.met.b* must be controlled for to partition *proximity.w* and *goal.met.w* into the appropriate within subject effects. The significance of the *proximity.b* and *goal.met.b* between-subjects effects will be examined in an exploratory manner.

The sjPlot (Lüdtke, 2025), merDeriv (Wang & Merkle, 2018), and interactions (Long, 2024) packages will be used for model summaries and calculating the cluster robust SEs and probing statistically significant interactions (if applicable). To test for parametric assumptions, we will use the performance (Lüdtke et al., 2021) and/or DHARMA (Hartig, 2024) packages.

Exploratory Analyses

Site level differences are not hypothesized. However, we will examine potential country/site differences in an exploratory manner. This will include exploring (a) whether Schweitzer et al.'s (2004) findings replicate within the United States (where the original study was conducted) and outside of it and (b) whether the findings are affected by the specific 90th

percentile goal used at each site.

When Schweitzer et al. (2024) discussed the potential for proximity effects in their data, they reported their results descriptively, by condition. Though this interaction was not tested, when participants were 1 word short of the goal, 14.81% of DYB, 34.48% of mere goal, and 48.48% of reward goal trials were overstatements. Though this proximity $\times$ group is worthwhile to investigate further, the potential for a three-way interaction with loss aversion could lead to a significant loss of power. Consequently, we plan to test the two- and three-way interactions in an exploratory manner, without a priori hypotheses.

Sample Size Adequacy and Power

Each of the eight main sites has committed to collecting 200 participants across the four conditions, which was determined to be feasible by each site lead. To ensure that the identified feasible sample size will provide sufficient power to detect meaningful effect sizes, we simulated data for a power analysis based on a variety of effect sizes smaller than or equal to the original effects in Schweitzer et al. (2004). Even though our statistical model will include site as a fixed effect predictor, we do not include site differences in the power analysis below because these differences are exploratory and we do not have any supporting information to reliably estimate differences between each site. To ground our power analysis, we focus on the smallest effect size reported by Schweitzer et al. (2004) – the effect of a “mere goal” on unethical behavior compared to a “do your best” goal – to provide a conservative power estimate. Because our statistical approach is different from Schweitzer et al.’s, we converted the Cohen’s d effect sizes from the original study into log odds (beta weights), then calculated power assuming that the true effect was equal to that found in Schweitzer et al. (2004). For more conservative estimates, we also calculated estimated power assuming the true effect is 75% and 50% of the original effect

sizes.

Table 2 shows the estimated power levels across effect sizes for each of the hypotheses. Our power analysis demonstrates that all hypotheses have > 80% power at both 75% and 100% of the smallest effect size reported by Schweitzer et al. (2004). At 50% of the original effect sizes, 2/5 hypotheses have 80% power. In two of these cases, the lower power involves contrasting the personal goal condition to others. Simulated power for this condition is difficult to estimate, due to it being a novel experimental condition with no prior data regarding the relevant incidence of overstatement. Further, if the differences between the groups are miniscule, it is unlikely to hold practical significance. In the third case, the effect of trait loss aversion had low statistical power at 50% effect size. However, as this effect was already estimated as being small (correlation of .15), half this effect would also point to it not being meaningful. Importantly, insufficient power for the effects at 50% size would not affect the replication aspect of this study. In summary, our power analyses suggest that we will have sufficient statistical power to appropriately test our hypotheses at the original effect sizes, with additional room for effect size shrinkage across almost all hypotheses.

Results

[As requested by the review team, we are providing the results section using simulated data until the data are collected.]

Sample Characteristics

Data was collected from XXX participants across X sites. Participants were excluded from the analyses if they (a) did not meet the inclusion criteria (i.e., enrollment in an English-taught university program; $n = X$), (b) did not participate in the task (i.e., skipped all rounds of the anagram task; $n = X$), and/or (c) indicated that their data should not be used ($n = X$).

Altogether, this resulted in XX participants being excluded from the analyses. The final sample comprised XX participants [description of demographics across and between sites].

At the trial level, data were excluded based on participant error (e.g., creating words from letters not in the list or not indicating whether goals were met). From the XX possible observations, XX were missing trial-level data (XX%). Due to using GLMMs with maximum likelihood estimation, missing data at the trial level can be treated as MAR and analyses can be conducted on all available data with minimal impact on power. In contrast, missing person-level data (i.e., trait loss aversion) requires listwise deletion for the relevant analyses. In total, XX participants did not answer the trait loss aversion questions.

Results from the manipulation check indicated that XX% of participants selected the response option corresponding to their condition. Following recent recommendations in the literature (e.g., Aronow et al., 2019; Mathur, 2025; Varaine, 2023), all participants were included in the analyses (i.e., regardless of their manipulation check results). The rationale for this approach is that unobserved variables could influence both comprehension/attention and our dependent variable (e.g., individual differences in conscientiousness could plausibly affect performance reporting and the likelihood of failing the manipulation check; in such cases, excluding participants based on the manipulation check could introduce selection bias rather than reduce noise). However, following our primary analyses, we also conducted supplementary analyses to explore how the results differ when those who failed the manipulation check are excluded.

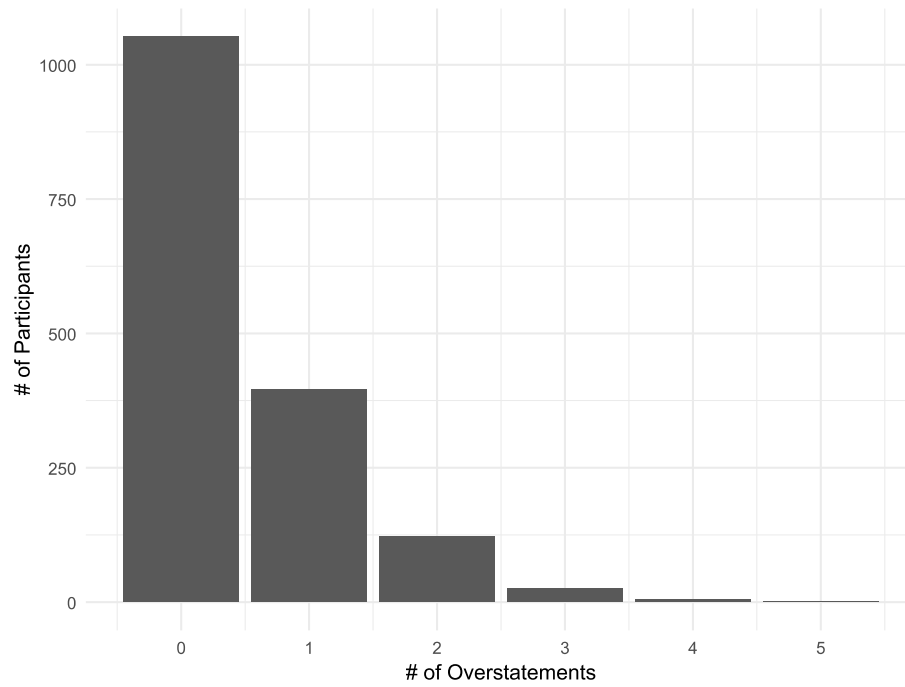


Figure 1. Frequency of Overstatements SIMULATED DATA

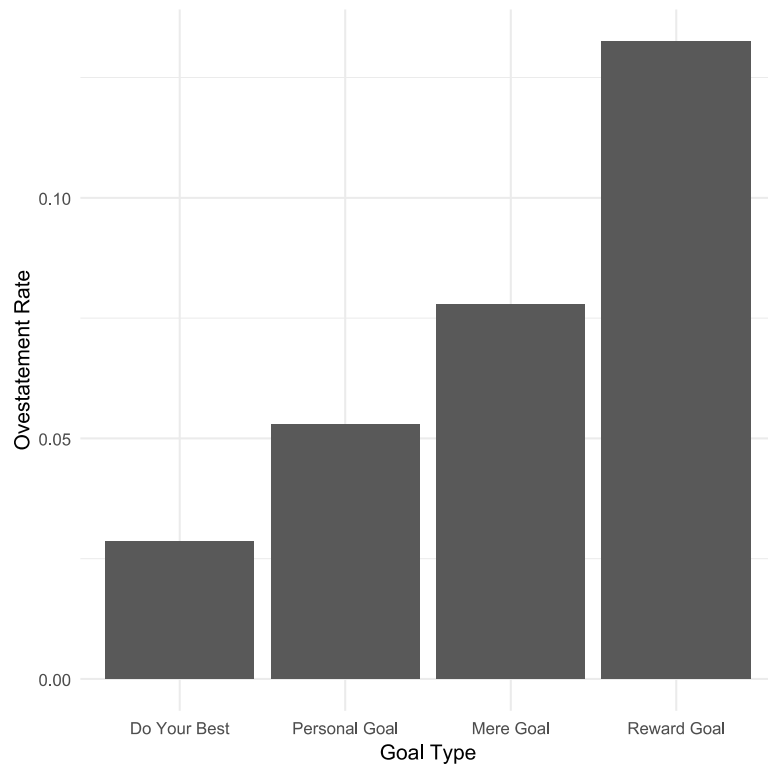


Figure 2. Overstatement rate by goal type SIMULATED DATA.

Descriptive Statistics

The mean number of completed words was XX ($SD = XX$) in the do-your-best condition, XX ($SD = XX$) in the personal goal condition, XX ($SD = XX$) in the mere goal condition, and XX ($SD = XX$) in the reward goal condition. Participants in the do-your-best condition met their goals XX% of the time, whereas personal goal participants met theirs XX% of the time, mere goal XX%, and reward goals XX%. XX% (site range = XX%-XX%) of participants overstated their number of completed words on the anagram task at least one time. The distribution of overstatements (by trial) skewed toward lower numbers (see Figure 1), with a mean of XX ($SD = XX$) and a median of XX. On average, participants in the do-your-best condition overstated XX times ($SD = XX$) over the 7 trials, in contrast to those in the personal goal ($M = XX$, $SD = XX$), mere goal ($M = XX$, $SD = XX$), and reward goal ($M = XX$, $SD = XX$) conditions. People who overstated more [were/were not] more likely to have high loss aversion at the zero-order level ($t(df) = XX$, $SE = XX$, $p = XX$).

Confirmatory Analyses

Six statistical models were tested (see Table 3 for a summary). Each model was examined for violations of statistical assumptions, including multivariate normality of random effects and overdispersion. Assumptions were [met/not met] for Models [unconditional/1/2/3/4/5], corresponding to hypotheses [1/2/3/4/5 [but not Models unconditional /1/2/3]]. [To account for the assumption violations... [autoregressive structures were added to the model/additional predictors were added/a beta-binomial distribution was used/AGQ integer was increased]]. The ICC_{uandj} for the unconditional model was XX, suggesting that between-subject effects can account for XX% of the variance in logit-linked overstating.

Hypothesis 1 examined evidence to support specific challenging goals leading to

overstating behavior. Specifically, participants from the personal goal, mere goal, and reward goal conditions were tested for higher overstatement rates than participants in the do-your-best condition. Results indicated that participants in the personal goal condition had a XX times [higher/lower] odds to overstate their performance than participants in the do-your-best condition. Participants in the mere goal condition had a XX times [higher/lower] odds of overstating their scores than participants in the do-your-best condition, and those in the reward goal condition had a XX times [higher/lower] odds of overstating their scores than participants in the do-your-best condition. Because [all/not all] group comparisons were statistically significant, we can conclude that goal-setting [was/was not] associated with significantly more overstating behavior than when people were told to simply do their best. Hypothesis 1 [was/was not] supported.

Hypothesis 2 stated that reward goals are associated with more overstating than mere goals (H2a), which in turn are associated with more overstating than personal goals (H2b). Results [supported/did not support/partially supported] this hypothesis (Table 3). [For participants with mere goals, the odds of overstatement were XX times higher than those with personal goals. For participants with reward goals, the odds were XX times [greater/lesser] to overstate compared to those with mere goals.] Hypothesis 2 [was/was not/was partially] supported.

Across groups and proximity levels, loss aversion [was/was not] a significant predictor of overstatement. This effect was [small/medium/large] and [positive/negative] such that greater loss aversion was associated with [more/less] overstating. Hypothesis 3 [was/was not] supported.

Hypothesis 4 examined the effect of goal proximity on overstating behavior. A participant who created more words than their average but did not reach their goal was

[not/more/less] likely to overstate on that trial than they would otherwise. Hypothesis 4 [was/was not] supported.

Hypothesis 5 tested the interaction between goal proximity and loss aversion. Note that loss aversion was standardized and proximity was recentered such that the intercept lay at participants being one word away from the goal. A participant with average loss aversion had a [significant/non-significant] effect of goal proximity [such that greater/lower goal proximity was associated with a greater likelihood of overstating]. On the other hand, when participants were one word away from their goal, [higher/lower] loss aversion was associated with [more/less] overstating. A [significant/non-significant] interaction effect suggested [a/no] conditional relationship between the variables. [Goal proximity [amplified/attenuated] the effect of loss aversion, such that when participants were nearer their goal, loss aversion was [more/less] strongly associated with their likelihood to overstate. When participants were [1/2/3/4/5/6/7/8, (depending on simple slopes)] words away from their goal, greater loss aversion [predicted/did not predict] [greater/lesser] overstating. When they were [1/2/3/4/5/6/7/8, (depending on simple slopes)] words from their goal, this effect was [weaker/stronger], and greater loss aversion [predicted/did not predict] [greater/lesser] overstating. Consequently, Hypothesis 5 [was/was not] supported.

Table 3. *Odds Ratios and 95% CIs for the Unconditional Model and Models 1-2 [using SIMULATED DATA WITH 10% MISSINGNESS COMPLETELY AT RANDOM]*

Predictor	Unconditional Model	Model 1	Model 2
Intercept	0.06*** [0.05, 0.06]	0.03*** [0.02, 0.03]	0.03*** [0.02, 0.03]
DYB vs Personal		1.83*** [1.35, 2.48]	1.83*** [1.35, 2.48]
DYB vs Mere		2.59*** [1.93, 3.46]	
DYB vs Reward		4.90*** [3.72, 6.48]	
Personal vs Mere			1.41** [1.10, 1.82]
Mere vs Reward			1.90*** [1.52, 2.36]
<i>N</i>	1600	1600	1600
Random intercept variance τ_{00}	1.26	0.51	0.51
ICC_{unadj}	.201	.123	.123
R^2_{cond}	.201	.203	.203

Notes. DYB = Do-your-best condition. ICC_{unadj} is the proportion of variance accounted for by random effects. R^2_{cond} is the proportion of variance accounted for by fixed and random effects. * $p < .05$, ** $p < .01$, *** $p < .001$.

Table 4. Odds Ratios and 95% CIs for Models 3-5 [using SIMULATED DATA WITH 10% MISSINGNESS COMPLETELY AT RANDOM]

Predictor	Model 3	Model 4	Model 5
Intercept	0.57*** [0.48, 0.67]	0.58*** [0.49, 0.68]	0.57*** [0.48, 0.67]
DYB vs Personal	2.17*** [1.51, 3.12]	2.13*** [1.49, 3.04]	2.18*** [1.51, 3.14]
Personal vs Mere	1.64** [1.20, 2.25]	1.72** [1.26, 2.34]	1.73** [1.26, 2.37]
Mere vs Reward	2.30*** [1.73, 3.05]	2.21*** [1.67, 2.91]	2.28*** [1.71, 3.02]
Proximity _{within}	6.02*** [5.18, 7.01]	6.04*** [5.19, 7.02]	6.12*** [5.25, 7.14]
Proximity _{between}	1.11 [0.93, 1.33]	1.09 [0.91, 1.29]	1.09 [0.91, 1.30]
Goal met _{within}	0.01*** [0.00, 0.02]	0.01*** [0.00, 0.02]	0.01*** [0.00, 0.02]
Goal met _{between}	0.88 [0.32, 2.38]	0.91 [0.34, 2.41]	0.90 [0.33, 2.44]
Loss Aversion (z-scored)		1.40*** [1.26, 1.55]	1.82*** [1.55, 2.13]
Loss Aversion × Proximity _{within}			1.44*** [1.27, 1.64]
Loss Aversion × Proximity _{between}			0.98 [0.82, 1.17]
Loss Aversion × Goal met _{within}			0.45** [0.25, 0.80]
Loss Aversion × Goal met _{between}			1.44 [0.54, 3.86]
<i>N</i>	1600	1600	1600
Random intercept variance τ_{00}	0.82	0.68	0.76
ICC_{unadj}	.054	.045	.049
R^2_{cond}	.782	.782	.789

Notes. ICC_{unadj} is the proportion of variance accounted for by random effects. R^2_{cond} is the proportion of variance accounted for by fixed and random effects. * $p < .05$, ** $p < .01$, *** $p < .001$.

[The supplementary analyses will follow the same general procedures as the primary analyses described above. Given that the specific nature of the supplementary data is difficult to anticipate (e.g., we do not know how many participants will fail attention checks), we will provide a Supplementary Results section at Stage 2, following the data collection. These results will include the exploratory analyses previously discussed]

Discussion

[Consistent with Psychological Science's guidelines for registered reports, we will provide an interpretation and discussion of the findings at Stage 2, following the data collection.]

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Appendix H

Psychological Science Stage 1 In-Principle Acceptance Letter

Decision Letter (PSCI-25-0190.R3)

From: yoel.inbar@utoronto.ca

To: christopher.castille@nicholls.edu

CC:

Subject: PSCI-25-0190.R3 - Decision on Stage 1 Manuscript

Body: 15-Apr-2026

Dear Dr. Castille:

Thank you for submitting your Stage 1 manuscript to Psychological Science for consideration as a Registered Report. We are pleased to offer in-principle acceptance of your submission, "Registered Report: Do Specific Challenging Goals Promote Unethical Behavior? Insights From a Global, Pre-Registered, Constructive Replication Study". You may now progress to Stage 2 and complete the study as approved. We would be grateful if you could now update the journal office as to the anticipated completion date of your study.

Following completion of your study, we invite you to resubmit your paper as a revision for peer review as a Stage 2 Registered Report. Please note that your manuscript can still be rejected for publication at Stage 2 if the Editors consider any of the following conditions to be met:

- The data do not pass the approved outcome-neutral criteria (such as absence of floor and ceiling effects or success of positive controls)
- The Introduction, rationale and stated hypotheses are different from those approved during the Stage 1 submission
- You do not adhere to the registered experimental procedures
- Unregistered exploratory analyses are not identified as exploratory, well justified, methodologically sound, and informative
- Your conclusions are not justified given the data

We encourage you to read the complete guidelines for authors concerning Stage 2 submissions at https://www.psychologicalscience.org/publications/psychological_science/ps-submissions. Please especially note the requirements for data sharing and that withdrawing your manuscript will result in publication of a Retracted Registration.

When you are ready to submit your Stage 2 Registered Report, log into <https://mc.manuscriptcentral.com/psci> and enter your Author Center, where you will find your report title listed under "Manuscripts with Decisions." Under "Actions," click on "Create a Revision." Your manuscript ID number will be appended to denote a revision.

IMPORTANT: Your pre-data manuscript files are available to you when you upload your revision. Please delete any redundant or outdated files before completing the submission of your post-data manuscript.

Once again, thank you for submitting your manuscript to Psychological Science. I look forward to receiving your post-data manuscript.

Sincerely,

Yoel Inbar
Senior Editor, Psychological Science

Reviewer comments:

Date Sent: 15-Apr-2026

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